REFERENCES

No	Document	Description
1	5.2.2.1 - Site Specific Information_Rev 1	Owner Growth Project - Process Plant - Site Specific Information
2	4.7.1 - Consultant Report 7555 - Owner PDC Data Summary Rev 0, March 2022	Table 2-5, "Back Calculated Ore Parameters", 2014 - 2021 value.
4	4.6.1 - Owner Growth Mass Balance_Nominal Case Report_Rev C	Mass Balance - SysCAD Report Nominal Case. 22/03/2022
5	Owner Comminution Plant Data_B.xlsx"	Benchmarking from FIM pebble crushing circuit.
7	ECI Tender from Consep	Q12375 - EPC Gravity Circuit Rev1
8	5.1.12f - BOD Alignment Template - Flotation Tail PLT, FT CIL, Elution & EW	Basis of Design for Flotation Tailings CIL, Elution & EW circuit. 06/04/2022
9	4.7.3 - Consultant Report 7555 Owner PDC PSD Addendum Rev 1	Owner - Process Design Criteria Summary - PSD Addendum Report No. 7555 Rev 1 March 2022
10	5.1.12g - BOD Alignment Template - Reagents and Services	5.1.12g - BOD Alignment Template - Reagents and Services - Rev C Owner confirmation that regrind mill ball size will be 25 mm (cell F15).
14	Owner Growth Project ECI Outcome Letter - EPC	Owner Growth Project - Process Plant - Formal Agreement for Engineering, Procurement and Construction: Early Contractor Involvement (Value Engineering Phase). 03/08/2022
16	Owner Process Design Criteria - VEP Phase (Stage 1)_I	Owner - Growth Project - Process Design Criteria Rev I. 20/09/2022
18	Cleaner Option 2- Owner Case 2 Scav Con to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner
19	Cleaner Scavenger - Owner Case 2 Scav to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner Scavenger
23	Owner GP Process Design Criteria - Reliance Information S1 Rev K, 25/01/2023	Principal supplied reliance information.
24	VEP Outcome Letters - EPC 09/02/23 and 02/05/23	Principal supplied Scope of Works letter for Stage 1 and 2
25	Consultant Report 7653 - Owner Value Engineering Update	Grinding circuit modelling for Owner by Orway Mineral Consultants, 160223
26	Process Design Workshop Meeting Minutes 170223, MOM-PM-SFP002	Workshop between the Client and EPC
27	Glencore Jameson Cell Sizings and Quote	Simulations from Vendor, 22/02/24 Owner Expansion Project - Jameson Cell Quote OTJ-2220 7532.3
28	Weir Cyclone Simulations and Datasheet - Primary & Gravity Cyclones	Various simulations for the different process cases, and datasheet: 2000-F00-DSH-ME-11005A_B Primary Cyclone
31	Trash Screen Spray Water Information from Vendor	Emails with Vendor, 16/08/23
32	Flotation Mass Balance Scenarios	Mass Balances for Process Margins Rev H and New Mass Balance Cases Rev G
33	Cadia Systems Gravity Electrowinning Cell	Emails with Vendor, 14/03/23. Ag and Cu Assays from site, provided by Owner (23/03/23)
35	Discussions with Client and TQ regarding Cyanide Destruction	Emails with Client, 230323 and TQ: 2000-F72-TQY-ME-10094_0_TQ-R
36	Vendor Information from Transmin for New Flocculant Mixing Plant	Transmin Proposal Execution Phase rev 3, P33-904-06
37	Direction from EPC Mechanical Engineers re: COS	Email, 280323
38	Client Information regarding Grinding Media	Emails with Client, 070323
39	Vendor Information from Consep for the Acacia CS8000 ILR	Emails with Client, 210322
40	EPC Calculation	SAG_Ball_Screen_Trommel_Calc_Rev_C.xlsx
41	Client Direction, Transfer of FLS Equipment Technical Information	Emails from client, various dates (May 2023)
42	Primary Crusher Datasheet	Scope of Work - Primary Crusher, Appendix A, doc 2000-F12-SOW-ME-20001 rev 1
43	SoW for Separable Portions Stage 1&2	Scope of Works between Owner and EPC 2000-F00-SOW-CO-20001 & 2000-F00-SOW-CO-20002_0
44	Outstanding Process Inputs from Owner	Outstanding Process Inputs 210623 (version 1) Additional (003) (002), and 2000-F00-TQY-PR-10051.0.TQ-R.0.01
45	Air Compressor Vendor Information	Atlas Copco Tender: 3151253_20220408045752 (1) rev 1- pdf copy
49	Owner email Owner GP - Float Cell Motors", dated 11 May 2023	FLSmidth flotation blower details.
50	Calculation for LPG Consumption and TPE ECI Tender	CAL-PR-005_A LPG Consumption TPE5000 ELUTION HEATER PACKAGE - ME16 TENDER - COMPLETE 20/05/22
51	FLSmidth - Primary Crusher Application Model	FLS_13x19 TSUV_Calc_FIM_Run 15_13x19 TSUV_ 6.5 OSS_3716 mtph_70_AVG results
53	EPC email Apron Feeder Datasheet Internal Review", dated 30 August 2023	Agreement with Owner to change contract SOW design factor from 120% to 110% of nominal tonnage rate.
54	Weir UFG Cyclone Cluster Modelling	UFG_CY_Select_Design_RevC UFG_CY_Select_Nominal_RevC
55	Takraf Tender Rev 0 Datasheet	2000-F00-DSH-ME-11018_B Trash Screens Completed
56	Owner GP Process Design Criteria - Reliance Information S1 Rev L, 25/01/2023	Principal supplied reliance information.
57	Owner Growth Project SysCAD Model (Historical)	SysCAD Model files and Reports (2022.08.31)-01.spf
58	NTF Deslime & UFG Cyclone PDC Inputs	Word document 260723
59	Joest Vibrating Screens Offer	JOEST_M22-11102 Rev D_JOEST FIRM Offer 2000-F00-DSH-ME-11011_C Vibrating Screens
60	Flotation Launder Wash Water Technical Query	Technical Query (2000-F16-TQY-ME-10007) FLS Launder Water Requirements Rev.0 31/07/2023
62	Technical Query - Design Bulk Density Range for Apron Feeders And Conveyors	2000-F00-TQY-ME-10034
63	Flotation Area - Design P80 Sizes	Email Correspondence. RE - Flotation D80 Mech Requests. 250823
64	Reagent Delivery Formats, 2000-F72-TQY-ME-20001_2	Reagent delivery information, provided by Owner
65	IsaMill Quote and Modelling by Glencore	IsaMill Quote OMT232007747 Owner - Owner EPC - IsaMill Sizing - Design -19102023 - New data supplied water temp 25 Owner - Owner EPC - IsaMill Sizing - Nominal -19102023 - New data supplied water temp 25 Email: "FW_ ME039 IsaMill Latest MB - Clarifications 64 - UFG Mill Media Hopper Live Volume"
66	Regrind Removal TQ	2000-F00-TQY-GE-10080_0 - RESPONSE.pdf
67	Client Comments Stage 1 Rev K PDC and Rev L SysCAD	2000-F00-DCR-PR-10001.K.ICR-R.K.02

REFERENCES

No	Document	Description
68	Raw Water TQ 10044	2000-F24-TQY-PR-10044.0.TQ-R.0.0164
69	SysCAD & PDC Client Comments TQ 10107 Response	2000-F00-TQY-PR-10107.0.TQ-R.0.02
70	Gravity Variation (VAR-017) and Communication from the Client	P3-EPC-COR-085 Variation Direction No.017 Gravity Circuit Modifications Emails from Client/Outcome of meetings
71	Emails from Consep (Knelson & ILR) in relation to Gravity Variation (VAR-017)	Email dated 30/10/23, 'RE: Knelson Concentrators (KC-QS70) and CS10000'
72	Gravity Variation TQ 10115 Response from Client	2000-F14-TQY-PR-10115.0.TQ-R.0.01 2000-F14-TQY-PR-10115_1_TQ-R
73	Quotation and emails from Cadia Systems (EW cells) in relation to Gravity Variation (VAR-017)	Q231115-PRI_EPC_EWC 151123 issue RE_ Owner - Gravity Pregnant Solution Electrowinning - various emails
74	Variation 018 UFG Mill 2 Trash Screen	P3-EPC-COR-104 Variation Direction No.018 UFG Mill 2 Trash Screen
75	Clarification Emails for Flotation Reagent Calculations	Emails with Client 21/11/23: 'RE: Owner GP - PAX Usage/Flow Consumption/Storage Tank RT'
76	Dump Pocket TQ from Owner	2000-F12-TQY-GE-10127_0_TQ-R
77	Wet Static Screens Datasheet (ME40) and Spray Water Information	2000-F00-DSH-ME-11040_1 and emails dated 23/01/24
78	Owner Owner's Manual CAT793F Owner	Owner's Manual CAT793F Owner
79	Water Services PFD Client Review	2000-F00-PFD-PR-10046.D.IFD-R.D.02
80	Area 30 M1 Design Review Report	2000-F30-REP-GE-10101_A - M1 Design Review Report
83	TQ147 Reagent Tank Volumes, Var 26 Cyanide Tanks and Var 192 CuSO4 Tanks	2000-F72-TQY-ME-10147_0_TQ-R P3-EPC-COR-162 Variation Direction No.026 Cyanide Tank Storage Capacity P3-EPC-COR-192 Variation Direction No.030 Copper Sulphate Storage
88	TQ206 Request for LPG & RTO Information	2000-F00-TQY-PR-10206.0.TQ-R.0.0270
90	Client Comments Stage 1 Rev 0 PDC	2000-F00-DCR-PR-10001.0.IFD-R.0.02
92	SAG & BM Trommel Sizes	2000-ME001.2-B02-50041_C SAG Mill Trommel Sizing Report 2000-ME001.2-M03-50001_B SAG Mill Trommel Assembly 2000-ME001.3-B02-50020_B Ball Mill Trommel Sizing Report 2000-ME001.3-M03-50001_B Ball Mill Trommel Assembly
100	5.1.1 - ECS Process Design Criteria	Engineer Process Design Criteria - Owner Growth Project ECS. 25/01/2022 J6027-000-DCR-PR-001 Rev D
101	5.1.2 - ECS Mass Balance	Engineer Mass Balance - Owner Growth Project ECS. 25/01/2022 J6027-000-CAL-PR-001 Rev D
102	4.5.1 - Process Design Criteria Rev F	Owner - Growth Project - Process Design Criteria Rev F. 16/03/2022
103	4.5.1 - Owner Process Design Criteria - VEP Phase _H, 19/08/2022	Principal supplied reliance information.
105	2000-F30-TQY-PR-15001.0.TQ-R.0.02	Client Response to Technical Query - CIL2/3 - Carbon Transfer Rate Between Adsorption Tanks
106	2000-F00-DCR-PR-10002.C.ICR.-R.C.02	Client comments to Process Design Criteria - Stage 2
111	Hydrogen Peroxide Technical Data Sheet	Solvay Hydrogen Peroxide Technical Datasheet
112	PAX, CuSO4 & Lead Nitrate Delivery Sizes	Email from IXOM (vendor), dated 28/03/2024
113	Concentrate Thickener Vendor GA	2000-ME091-B02-10001_C

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
1.0	**PRODUCTION SCHEDULE**												
1.1	**Throughput**												
	Annual Throughput		t/a	10,500,000	25,500,000	23,997,254	25,500,000	28,350,000	1	Owner	16, 24 & 26		I
	Annual Operating Days		d/a	365	365	365	365	365	1	Owner	16		E
1.2	**Operating Schedules**												
	Crushing												
	Plant Availability		%	65.0	65.0	65.0	65.0	65.0	1	Owner	1, 30		H
	Annual Operating Hours		h/a	5,694	5,694	5,694	5,694	5,694	6	Calculated			
	Hourly Throughput - Existing Circuit		t/h	1,844	1,844	1,844	1,844	1,844	6	Calculated			
	Hourly Throughput - New Circuit		t/h		2,634	2,370	2,634	3,135	6	Calculated			
	Total Hourly Throughput		t/h	1,844	4,478	4,214	4,478	4,979	6	Calculated			
	Concentrator												
	Plant Availability		%	91.3	93.4	93.4	93.4	93.4	1	Owner	16, 23		G
	Annual Operating Hours - Existing SAG Circuit		h/a	8,000	-	-	-	-	6	Calculated			
	Annual Operating Hours - New SAG/Ball Mill Circuit		h/a	-	8,182	8,182	8,182	8,182	6	Calculated			
	Hourly Throughput - Existing SAG Circuit		t/h	1,313	1,313	1,313	1,313	1,313	6	Calculated			
	Hourly Throughput - New SAG/Ball Mill Circuit		t/h		1,833	1,650	1,833	2,182	6	Calculated			
	Total Hourly Throughput		t/h	1,313	3,146	2,962	3,146	3,494	6	Calculated			
	UFG Circuit												
	Plant Availability		%	91.3	93.4	93.4	93.4	93.4	1	Owner	16, 23		G
	Annual Operating Hours		h/a	8,000	8,182	8,182	8,182	8,182	6	Calculated			
2.0	**MATERIAL CHARACTERISTICS**												
2.1	**General Ore Properties**												
	Moisture Content		%w/w			1.3			1	Owner	16		A
	Ore Specific Gravity	Nominal				2.90			1	Owner	16		A
		Range				2.70 - 3.10			1	Owner	23		K
	Ore Bulk Density	Normal				1.90			1	Owner	16		A
		Range				1.70 - 2.10			1	Owner	56		K
		Design Range				1.60 - 1.90			1	Owner	62	To be used for all conveyors & apron feeders in the crushing and milling circuits	K
	Angle of Repose		degrees			36			1	Owner	16		A
	Drawdown Angle		degrees			60			1	Owner	16		A
2.2	**Ore Comminution Parameters**												
	Crushing Work Index												
	Testwork Average		kWh/t			17.8			1	Owner	16		A
	Testwork Range		kWh/t			16 - 30			1	Owner	16		A
	Testwork 80th Percentile		kWh/t			18.8			1	Owner	2, 23	2014 - 2021 value (as of rev E)	G
	Abrasion Index												
	Testwork Average					0.145			1	Owner	16		A
	Testwork Range					0.020 - 0.680			1	Owner	16		A
	Testwork 80th Percentile					0.192			1	Owner	16		A
	Ball Mill Work Index	Range	kWh/t			11.5 - 15			1	Owner	41		K
		Design	kWh/t			12.5			1	Owner	2	2014 - 2021 value	E
	A*b	Range				38 - 48			1	Owner	41		K
		Design				43.8			1	Owner	2	2014 - 2021 value	E
2.3	**Pebble Comminution Parameters**												
	Crushing Work Index	Design	kWh/t			18.8			1	Owner	16		E
		Maximum	kWh/t			23.0			1	Owner	23		K
	Bulk Density	Minimum	t/m^3			1.60			1	Owner	16, 62		K
		Maximum	t/m^3			1.90			1	Owner	41, 62		K
2.4	**Nom Ore Grades**												
	Feed Grade - S		% S	1.02	1.02		1.40	1.02	1	Owner	23	Achievable grade back calculated by EPC (Source 32)	I
	Feed Grade - Au		Au (g/t)	1.58	1.58		1.70	1.58	1	Owner	23	Achievable grade back calculated by EPC (Source 32)	I
	Final Tail Grade - S		% S	0.18	0.65		0.72	0.78	1	Owner	23	Blue data is from Owner Ref 23	G
	Final Tail Grade - Au		Au (g/t)	0.23	0.39		0.40	0.41	1	Owner	23	Blue data is from Owner Ref 23	G
2.5	**Design Ore Grades**												
	Feed Grade - S		%S	1.49		1.49			1	Owner	23		G
	Feed Grade - Au		Au (g/t)	1.81		1.81			1	Owner	23		G
	Final Tail Grade - S		% S	0.29		0.77			1	Owner	23	Blue data is from Owner Ref 23	G
	Final Tail Grade - Au		Au (g/t)	0.38		0.42			1	Owner	23	Blue data is from Owner Ref 23	G
2.6	**Recovery**												
	Total Gold Recovery	Nominal	%			75.60			6	Calculated		Relative to feed grade	
	Total Sulphur Recovery	Nominal	%			36.29			6	Calculated		Relative to feed grade	
	Gravity Gold Recovery	Nominal	%			14.00			1	Owner	23	Relative to feed grade	G
	Gravity Gold Recovery	Maximum	%			20.00			1	Owner	23	Relative to feed grade	G
	Flotation Gold Recovery	Nominal	%			71.80			1	Owner	23	Relative to feed grade	G
	Flotation Gold Recovery	Maximum	%			75.00			1	Owner	23	Relative to feed grade	G
	Flotation Sulphur Recovery	Nominal	%			85.10			1	Owner	23	Relative to feed grade	G
	Flotation Sulphur Recovery	Maximum	%			90.00			1	Owner	23	Relative to feed grade	G
	Flotation Concentrate Leach Gold Recovery	Nominal	%			84.00			1	Owner	44	Range 80-88%	0
	Flotation Concentrate Leach Gold Recovery	Maximum	%			88.00			1	Owner	44	Range 80-88%	0
	Flotation Tails Leach Gold Recovery	Nominal	%			34.00			1	Owner	44	34% YTD	0
	Flotation Tails Leach Gold Recovery	Maximum	%			34.00			1	Owner	44	34% YTD	0
2.7	**Gold Production**												
	Nominal		oz Au/a			979,341			6	Calculated			
2.8	**Flotation Concentrate**												
	Production - Nom Grade		t/a			636,922			6	Calculated		25.5Mtpa and Nominal Grade, Case 1	
	Production - Design Grade		t/a			875,959			6	Calculated		24Mtpa and Design Grade, Case 2	
2.9	**Process Water SG**					1.070			1	Owner	1		E
2.10	**Scheme Water SG**					1.000			1	Owner	1		0
3.0	**CRUSHING CIRCUIT**												
3.1	**Circuit Parameters**												
	Throughput	Nominal	t/h	1,844	2,634	2,370	2,634	3,135	6	Calculated		New circuit	
		Design	t/h		3,135	3,135	3,135	3,135	1	Owner	43		K
		Maximum	t/h		3,436	3,436	3,436	3,436	1	Owner	56	Crusher fall through from FLS and not the circuit design basis	K
	Feed PSD												
	F_{99}		mm		1,000	1,000	1,000	1,000	1	Owner	23		K
	F_{80}		mm		450	450	450	450	1	Owner	23		K
	F_{50}		mm		150	150	150	150	1	Owner	23		K
	F_{20}		mm		25	25	25	25	1	Owner	23		K
	Product PSD												
	P_{80} - Maximum		mm	100	130	130	130	145	1, 5	Owner, Vendor	23, 25		I
3.2	**Dump Pocket**												
	Dump Pocket Type				Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	1	Owner	16		A
	Truck Model				793F	793F	793F	793F	1	Owner	23		G
	Truck Payload		t		250	250	250	250	1	Owner	78	Owner's Manual CAT793F.	0
	SAE 2:1 Heaped Tray Capacity		m^3		179	179	179	179	1	Owner	76, 78	Owner's Manual CAT793F.	0
	Dump Pocket Live Capacity		m^3		179	179	179	179	1	Owner	76, 78	Live capacity from one side	0
3.3	**Primary Crusher**												
	Type			Gyratory	Gyratory	Gyratory	Gyratory	Gyratory	1	Owner	16	Owner Supplied	A
	Model			54-75Mk II	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1	Owner	41	Owner Supplied	I
	Open Side Setting		mm	141	165	165	165	165	5	Vendor	51		H
	Throw		mm		38	38	38	38	5	Vendor	42		K
	Motor Power		kW	450	600	600	600	600	5	Vendor	42		K
	Simulation Results												
	P_{80}	Max	mm		122	122	122	122	5	Vendor	51	Simulated, see reference document for inputs	0
	P_{100}	Max	mm		229	229	229	229	5	Vendor	51	Simulated, see reference document for inputs	0

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
3.4	**Crushed Ore Storage**												
	Type			Stockpile	Stockpile	Stockpile	Stockpile	Stockpile	1	Owner	16		A
	Live Capacity - Existing		t	21,001	-	-	-	-	6	Calculated			
	Residence Time - Existing		h	16	-	-	-	-	1	Owner	16		K
	Bulk Density		t/m^3		1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1	Owner	43		K
	Live Capacity - New	using 60° draw down	m^3		12,035	12,035	12,035	12,035	2	EPC	37		H
	Residence Time - Minimum/Range	using 60° draw down	h		11.16	12.40	11.16	9.38	6	Calculated			
	Live Capacity at 1.70 t/m^3	using 60° draw down	t		20,460	20,460	20,460	20,460	6	Calculated			
	Live Capacity at 1.90 t/m3	using 60° draw down	t		22,867	22,867	22,867	22,867	6	Calculated			
	Live Capacity - New	using TUNRA data	m^3		11,293	11,293	11,293	11,293	2	EPC	37		H
	Residence Time - Minimum/Range	using TUNRA data	h		10.47	11.64	10.47	8.80	6	Calculated			
	Live Capacity at 1.70 t/m^3	using TUNRA data	t		19,198	19,198	19,198	19,198	6	Calculated			
	Live Capacity at 1.90 t/m3	using TUNRA data	t		21,457	21,457	21,457	21,457	6	Calculated			
	Primary Crusher Discharge Feeder												
	Number				1	1	1	1	1	Owner	43		K
	Type				Apron	Apron	Apron	Apron	1	Owner	16		K
	Capacity	Nominal	t/h		2,634	2,370	2,634	3,135	6	Calculated			
		Design	t/h		3,135	3,135	3,135	3,135	1	Owner	43		K
3.5	**Reclaim Feeders**												
	Number			4	2	2	2	2	1, 2	EPC Owner	16, 26		E
	Type			Apron	Apron	Apron	Apron	Apron	1	Owner	16		A
	Capacity per feeder	Nominal	t/h		917	825	917	1,091	6	Calculated			
		Design	t/h		2,017	2,017	2,017	2,017	9	Agreed	53	110% of nominal tonnage rate.	K
3.6	**Emergency Feeders**												
	Number				1	1	1	1	1	Owner	16		K
	Type				Apron	Apron	Apron	Apron	1	Owner	16		A
	Capacity per feeder	Nominal	t/h		1,833	1,650	1,833	2,182	6	Calculated			
		Design	t/h		2,017	2,017	2,017	2,017	9	Agreed	53	110% of nominal tonnage rate.	K
	Loader Model				992K/994K	992K/994K	992K/994K	992K/994K	1	Owner	23		G
4.0	**GRINDING AND CLASSIFICATION**												
4.1	**Circuit Parameters**												
	Type			SABC	SABC	SABC	SABC	SABC	1	Owner	16		A
	Throughput	Nominal @ 180 µm	t/h	1,313	1,833	1,650	1,833		6	Calculated			
		Nominal @ 220 µm	t/h					2,182	6	Calculated			
	Bulk Density	Design Range	t/m^3		1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1	Owner	43, 62		K
	Milling Circuit PSD												
	F_{80}		mm	100	130	130	130	145	1, 5	Owner, Vendor	23, 25		H
	P_{80}	Target	µm	180	180	180	180	220	1	Owner	16, 23, 24		H
	SAG Mill Specific Energy - 50th Percentile		kWh/t	8.3					1	Owner	16		E
	SAG Mill Specific Energy - 80th Percentile		kWh/t	8.9					1	Owner	16		E
	Ball Mill Specific Energy - 50th Percentile		kWh/t	5.4					1	Owner	16		E
	Ball Mill Specific Energy - 80th Percentile		kWh/t	6.1					1	Owner	16		E
	P_{80}	Modelled	µm		189	189	189	220	1	Owner	25		H
	SAG Mill Specific Energy Modelled		kWh/t		9.14	9.14	9.14	7.68	1	Owner	25		I
	Ball Mill Specific Energy Modelled	@ 180 µm	kWh/t		9.68	9.68	9.68		1	Owner	25		I
		@ 220 µm	kWh/t					8.14	2	EPC	25		I
4.2	**Primary Mill**												
	Type			SAG Mill	SAG Mill	SAG Mill	SAG Mill	SAG Mill	1	Owner	16	Owner Supplied	A
	Mill Diameter		m	10.97	10.97	10.97	10.97	10.97	1	Owner	16	Owner Supplied	A
	Effective Grinding Length - EGL		m	4.27	8.38	8.38	8.38	8.38	1	Owner	16, 23	Owner Supplied	G
	Duty Mill Speed Range		% Cs	60 - 80	60 - 83	60 - 83	60 - 83	60 - 83	1	Owner	24		H
	Duty Ball Charge		% v/v	13 - 17	10.5	10.5	10.5	10.9 - 11.3	1	Owner	16, 24		G
	Duty Total Load		% v/v	25 - 30	27	27	27	25	1	Owner	16, 24		G
	Total Ball Charge	Structural	% v/v		18	18	18	18	1	Owner	23	For structural design	G
	Total Load	Structural	% v/v		35	35	35	35	1	Owner	23	For structural design	G
	Mill Discharge Density		% w/w	75	75	75	75	75	1	Owner	9, 69		A
	Mill Pinion Power	Average	kW	10,894	-	-	-	-	1	Owner	25		E
	Mill Pinion Power	Duty	kW	11,682	16,757	15,078	16,757	16,755	1	Owner	25		H
	Installed Motor Power		kW	12,000	20,000	20,000	20,000	20,000	1	Owner	41, 23		G
	Liner Type				Steel	Steel	Steel	Steel	1	Owner	41		K
	Curved Pulp Lifter Capacity	Max	m^3/hr		2,100	2,100	2,100	2,100	1	Owner	41		K
	Pebble Port Aperture		mm		75	75	75	75	1	Owner	41	FLS offer	I
	Discharge Grate Open Area		%		11 - 12	11 - 12	11 - 12	11 - 12	1	Owner	41		K
	Trommel Dimensions	ID x L	m		3.829 m dia. X 3.80 m	3.829 m dia. X 3.80 m	3.829 m dia. X 3.80 m	3.829 m dia. X 3.80 m	5	Vendor	92		1
	Trommel Open Area		%		25	25	25	25	1	Owner	41		K
	Trommel Aperture		mm		12 x 40	12 x 40	12 x 40	12 x 40	1	Owner	41		K
	Spray Water		m³/h	150	150	150	150	150	2	EPC	40		K
	Trommel Undersize		t/h		1050	1050	1050	1050	2	EPC	32		K
	Existing Fim SAG Mill Transfer Density		% w/w	60					1	Owner	57, 69		A
4.3	**Primary Mill Discharge Screen**												
	Discharge Screen Type				Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	1	Owner	16		G
	Screen Size	W x L	m		4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	5	Vendor	59		0
	Effective Screen Area		m^2		36.47	36.47	36.47	36.47	5	Vendor	59		0
	Screen Open Area		%		34.0	34.0	34.0	34.0	5	Vendor	59		0
	Screen Aperture		mm		12 x 55	12 x 55	12 x 55	12 x 55	2	EPC	40		H
	Specific Screen Flux Rate		m^3/h/m^2 OA		18.68	18.68	18.68	29.90	5	Vendor	59	Includes wash water	0
	Oversize Moisture		%w/w	5	13	13	13	13	5	Vendor	59		0
	Spray/Wash Water		m³/h		197.2	197.2	197.2	197.2	5	Vendor	59		0
	Discharge Screen Feed % Solids		%w/w		85	85	85	85	2	EPC	32		K
	Discharge Screen Oversize		t/h		305	305	305	610	1	Owner	41		K
4.4	**Secondary Mill**												
	Type				Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	1	Owner	41	Owner Supplied	K
	Mill Diameter		m		8.53	8.53	8.53	8.53	1	Owner	23	Owner Supplied	K
	Effective Grinding Length - EGL		m		13.26	13.26	13.26	13.26	1	Owner	23	Owner Supplied	G
	Duty Mill Speed		% Cs		60 - 78	60 - 78	60 - 78	60 - 78	1	Owner	41		K
	Duty Ball Charge	Nominal	% v/v		28.5	28.5	28.5	28.5	1	Owner	41		K
	Duty Ball Charge	Maximum	% v/v		31.0	31.0	31.0	31.0	1	Owner	24		G
	Total Ball Charge	Structural	% v/v		38	38	38	38	1	Owner	23	With polymet liners. For structural design	G
	Mill Discharge Density		% w/w		75	75	75	75	1	Owner	4		E
	Mill Pinion Power	Duty	kW		17,747	15,969	17,747	17,759	6	Calculated			
	Installed Motor Power		kW		20,000	20,000	20,000	20,000	1	Owner	23		G
	Liner Type				Poly-Met	Poly-Met	Poly-Met	Poly-Met	1	Owner	23		G
	Trommel Flux Rate	Nominal	m^3/h/m^2		230	223	229	226	6	Calculated			
	Trommel Dimensions	ID x L	m x m		4.992 x 5.80	4.992 x 5.80	4.992 x 5.80	4.992 x 5.80	5	Vendor	92	ID = internal diameter	1
	Trommel Capacity	Maximum	m^3/hr		6,681	6,681	6,681	6,681	1	Owner	23	Includes spray water	G
	Mill Discharge Flowrate	Nominal	m^3/hr		6,378	6,203	6,368	6,282	6	Calculated		Includes spray water	
	Trommel Open Area	Minimum	%		30	30	30	30	1	Owner	23		G
	Trommel Aperture		mm		19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	1	Owner	23		G
	Trommel Screen Spray Water		m³/h		150	150	150	150	1	Owner	4		A
4.5	**Pebble Crusher**												
	Type			Cone	Cone	Cone	Cone	Cone	1	Owner	16	Owner Supplied	E
	Number			2 (duty / standby)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	1	Owner	16, 23	Owner Supplied	E
	Model			Symons 7ft	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	1	Owner	41	Owner Supplied	K
	Installed Motor Power per Crusher		kW	375 / 260	375	375	375	375	1	Owner	16, 41, 44		K
	Closed Side Setting		mm	7 - 9	7	7	7	7	1	Owner	16,41		K
	Throughput	Nominal	t/h	302					6	Calculated			
		Nominal	t/h		305	305	305	610	1	Owner	41		K
		Range	% of new feed	23 - 27					1	Owner	16		K
			% of new feed	23					1	Owner	16		K
			% of new feed		16.6	18.5	16.6	28.0	6	Calculated			
	Crusher Capacity at CSS	Maximum	t/h		610	610	610	610	1	Owner	41	305 t/h limit per crusher unit at CSS. This throughput represents the increased pebble load as the pebble ports wear out on SAG mill discharge grate	K
	Feed Bin Capacity	per bin	min	50	12.8	12.8	12.8	6	1,2	Owner, EPC	26		G
		per bin	t	250	65	65	65	65	6	Calculated		Total Bin capacity of 130 t (2 bins)	
		per bin	m^3	132	34	34	34	34	6	Calculated		Total Bin capacity of 68 m^3 (2 bins)	
	Feed Moisture	Range	wt%		2 - 8	2 - 8	2 - 8	2 - 8	1	Owner	41		K
	Feed Pebble PSD												
	F_{100}		mm		95.0	95.0	95.0	95.0	1	Owner	41	Linear interpolation from PSD data	K
	F_{80}		mm		49.0	49.0	49.0	49.0	1	Owner	41	Linear interpolation from PSD data	K
	F_{50}		mm		37.0	37.0	37.0	37.0	1	Owner	41	Linear interpolation from PSD data	K
	Product Pebble PSD												
	P_{80}		mm		12.0	12.0	12.0	12.0	1	Owner	69	FLS guaranteed value	0

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
4.6	**Classification**												
	Feed Size F_{50}	Typical	µm		276	276	276	276	1	Owner	16		E
	Feed Size F_{100}	Modelled	mm		350	350	350	350	1	Owner	9	Consultant Modelling	K
	Feed Size F_{80}	Modelled	mm		0.757	0.757	0.757	0.757	1	Owner	9	Consultant Modelling	K
	Feed Size F_{50}	Modelled	mm		0.306	0.306	0.306	0.306	1	Owner	9	Consultant Modelling	K
	Product Size P_{80}	Target	µm		180	180	180	220	1	Owner	26		H
	Product Size P_{80}	Maximum	µm		250	250	250	250	1	Owner	26		G
	Type				Hydrocyclone	Hydrocyclone	Hydrocyclone	Hydrocyclone	1	Owner	16		K
	Cyclone Model				800CVX13	800CVX13	800CVX13	800CVX13	5	Vendor	28		G
	Number of Cyclone Clusters				1	1	1	1	5	Vendor	28		G
	Distributor Outlets per Cluster		No.		18	18	18	18	5	Vendor	28		0
	Number Installed Cyclones		No.		17	17	17	17	5	Vendor	28	15 for primary cyclone duty, 2 for gravity duty	I
	Number Spare Outlets		No.		1	1	1	1	6	Calculated		1 blanked nozzle for the ladder access, 1 spare blanked nozzle for future	
	Number Operating Cyclones - Primary Cluster	Nominal	No.		12	11	12	12	5	Vendor	28		I
		Maximum	No.		14	14	14	14	5	Vendor	28		I
		Standby	No.		1	1	1	1	5	Vendor	28		I
		Total Installed	No.		15	15	15	15	6	Calculated			
	Number Operating Cyclones - Gravity	Nominal	No.		2	2	2	2	5	Vendor	28	Gravity cyclones require larger spigot and VF to achieve mass flow of 1000tph to underflow. There is also a higher recirc. load.	G
		Maximum	No.		2	2	2	2	5	Vendor	28		G
	Cyclone Feed	Nominal	t/h		14,420	13,986	14,420	14,623	6	Calculated			
	Cyclone Feed Flow	Nominal	m³/h		13,248	12,833	13,233	12,416	6	Calculated			
	Cyclone Feed Flow	Maximum @ 61.6%	m³/h		13,862	13,862	13,862	13,862	1	Owner	26	Flow rate of 13,862m3/hr requires 14 or 15 online, 2 gravity as advised by Weir	G
	Cyclone Feed Density		% w/w		62.0	62.0	62.0	65.0	5	Vendor	28		G
	Underflow Solids Concentration	Minimum	% w/w		75.0	75.0	75.0	75.0	1	Owner	16		K
	Underflow Solids Concentration		% w/w		77.2	77.0	77.2	77.2	6	Calculated			
	Underflow Solids		t/h		11,275	11,024	11,275	11,129	6	Calculated			
	Overflow Solids Concentration		% w/w		36.4	35.9	36.4	43.3	5	Vendor	28		G
	Overflow Solids		t/h		3,146	2,962	3,146	3,494	6	Calculated			
	Circulating Load - Cyclone UF / Cyclone OF		%		358	372	358	318	5	Vendor	28		K
	Circulating Load - Ball Mill Discharge / Cyclone OF		%		301	311	301	267	6	Calculated			
	Operating Pressure - Cyclones in Cluster		kPa		82	91	82	73	5	Vendor	28		K
	Operating Pressure - Gravity Cyclones		kPa		83	86	83	70	5	Vendor	28		K
4.7	**Trash Screen**												
	Type				Linear Screen	Linear Screen	Linear Screen	Linear Screen	1	Owner	16		A
	Total Number Screens				4	4	4	4	6	Calculated			
	Number Duty				3	3	3	3	1	Owner	14		F
	Number Standby				1	1	1	1	1	Owner	14		E
	Screen Dimensions	W x L	m		9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	5	Vendor	55		0
	Screen Area per Screen	Effective	m^2		40.03	40.03	40.03	40.03	5	Vendor	55		0
	Total Screen Area		m^2		120.09	120.09	120.09	120.09	6	Calculated			
	Effective Screen Area		%		34	34	34	34	5	Vendor	55		0
	Screen Aperture		mm		1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	5	Vendor	55		0
	Screen Feed Flow (incl. spray water)	per screen	m^3/h		2,187	2,097	2,186	1,939	6	Calculated			
	Maximum Screen Flux Rate		$m^3/m^2/h$		75	75	75	75	1	Owner	26		G
	Specific Screen Flux Rate		$m^3/m^2/h$		54.6	52.4	54.6	48.4	6	Calculated			
	Spray/Wash Water	per screen	m^3/h		60	60	60	60	5	Vendor	31, 55		K
	Total Spray/Wash Water	Total	m^3/h		180	180	180	180	6	Calculated			
	Spray Water reporting to Screen Oversize	per screen	m^3/h		45	45	45	45	5	Vendor	31		K
	Total Spray Water reporting to Screen Oversize	Total			135	135	135	135	6	Calculated			
	Spray Water reporting to Screen Undersize	per screen	m^3/h		15	15	15	15	5	Vendor	31		K
	Spray Water reporting to Screen Undersize	Total			45	45	45	45	6	Calculated			
	Water to Trash Screens Final Collection Box		m^3/h		258.0	134.9	258.0	306.8	6	Calculated		HP Process Water, back calculated in SysCAD	
4.8	**Trash Dewatering**												
	Dewatering Screen Type				Vibrating Screen	Vibrating Screen	Vibrating Screen	Vibrating Screen	1	Owner	16		K
	Number				1	1	1	1	1	Owner	16		E
	Screen Size	W x L	m		1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	5	Vendor	59		0
	Effective Screening Area per Screen		m^2		6.14	6.14	6.14	6.14	5	Vendor	59		0
	Specific Screen Flux Rate	Design	$m^3/m^2/h$		50	50	50	50	1	Owner	16		A
		Nominal	$m^3/m^2/h$		22.36	22.36	22.36	22.36	5	Vendor	59	Includes wash water.	0
	Screen Aperture		mm		1 x 15	1 x 15	1 x 15	1 x 15	5	Vendor	59		0
	Deck Type				Polyurethane	Polyurethane	Polyurethane	Polyurethane	5	Vendor	59		A
	Quantity of Wood/Grit/Trash removed per day		t/day		27	27	27	27	1	Owner	16		K
	Total Spray/Wash Water		m^3/h		11	11	11	11	5	Vendor	59		0
5.0	**GRAVITY CONCENTRATION**												
5.1	**Gravity Scalping Screen**												
	Type				Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	5	Vendor	59		A
	Number of Screens				2	2	2	2	1	Owner	16	Only 2x gravity concentrator circuits, no future allowance for additional circuits	A
	Screen Solids Feed Rate per Screen	Nominal	t/h		1,001	1,000	1,000	1,000	6	Calculated			
		Design	t/h		1,111	1,111	1,111	1,111	6	Calculated		To achieve 1,000 t/h to gravity concentrators at design O/S split	E
	Screen Feed Density	Target	% w/w		60.0	60.0	60.0	60.0	1	Owner	70	Excludes Spray Water	0
	Screen Feed Density	Actual	% w/w		60.0	60.0	60.0	60.0	6	Calculated			
	Screen Feed Flowrate per Screen	Nominal	m^3/h		969	967	968	968	6	Calculated		Excludes Spray Water	
		Design	m^3/h		1076	1075	1075	1076	6	Calculated		Excludes Spray Water	
	Screen Size	W x L	m		3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	1	Owner	59, 70, 72		0
	Effective Screening Area per Screen	Overall	m^2		22.31	22.31	22.31	22.31	5	Vendor	59		0
	Specific Screen Flux Rate	Nominal	$m^3/m^2/h$		48.0	47.9	48.0	48.0	6	Calculated		Includes Spray Water	
		Design	$m^3/m^2/h$		52.8	52.7	52.8	52.8	6	Calculated		Includes Spray Water	
		Target	$m^3/m^2/h$		50	50	50	50	1	Owner	16, 70		E
	Screen Split to Oversize - Range		%		8 - 15	8 - 15	8 - 15	8 - 15	1	Owner	16		E
	Screen Split to Oversize	Design	%		10	10	10	10	1	Owner	16, 72		0
	Screen Aperture		mm		3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	5	Vendor	59	3.2mm Slotted first half, 2mm non-pegging second half (VR) requested by Owner in source 16	0
	Screen Spray Water	per screen	m³/h		102	102	102	102	5	Vendor	59		0
	Oversize Solids Concentration		% w/w		87	87	87	87	5	Vendor	59		0
	Oversize Slurry Destination				Ball mill feed	Ball mill feed	Ball mill feed	Ball mill feed	1	Owner	43		A
	Undersize Slurry Destination				Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	1	Owner	43		A
5.2	**Gravity Concentrator**												
	Type				Knelson	Knelson	Knelson	Knelson	1	Owner	16		A
	Number of Concentrators				2	2	2	2	1	Owner	16		A
	Model				KC-QS70	KC-QS70	KC-QS70	KC-QS70	1	Owner	16		C
	Feed Capacity per Unit	Max	t/h		1,000	1,000	1,000	1,000	5	Vendor	39, 71		K
	Volumetric Capacity per Unit	Max	m³/h		910	910	910	910	5	Vendor	7		K
	Feed Solids Rate to Circuit	Nominal	t/h		1,800	1,800	1,800	1,800	2, 5	EPC Vendor	7, 71	300-1000t/hr/unit operating range	E
		Max	t/h		2000	2000	2000	2000	6	Calculated			E
	Feed Solids Rate per Unit	Nominal	t/h		900	900	900	900	6	Calculated		Vendor recommends 90% of the max (Source 71).	
	Feed Density		% w/w		54.2	54.2	54.2	54.2	6	Calculated			
	Volumetric Flowrate per Unit		m³/h		1022	1021	1021	1022	6	Calculated			
	Fluidising Water per Unit		m³/h		120	120	120	120	5	Vendor	7		K
	Total Fluidising Water	Nominal	m³/h		240	240	240	240	6	Calculated		120m3 per concentrator	
	Cycle Time (Inclusive of Flush Cycle)		min		30	30	30	30	5	Vendor	71		0
	Cycles Per Day		No.		48	48	48	48	6	Calculated			
	Concentrate Production Per Cycle		kg		150	150	150	150	5	Vendor	7, 71		E
	Concentrate Bulk SG		t/m^3		2.20	2.20	2.20	2.20	1	Owner	44		0
	Daily Concentrate Production		kg/d		14,400	14,400	14,400	14,400	6	Calculated			
			m³/d		6.55	6.55	6.55	6.55	6	Calculated			
	Fresh Feed to Concentrators		%		57.2	60.8	57.2	51.5	6	Calculated		As a % of fresh mill feed	
	Tailings Destination				Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	1	Owner	43		A
6.0	**INTENSIVE LEACH**												
6.1	**Intensive Leach Reactor**												
	Type				Acacia	Acacia	Acacia	Acacia	1	Owner	16		A
	Model				CS10000	CS10000	CS10000	CS10000	1	Owner	70		0
	Number				1	1	1	1	1	Owner	16		A
	Batches per Day	Nominal			1	1	1	1	5	Vendor	39		E
		Max			2	2	2	2	5	Vendor	39	Required to complete 2 batches per day as per Owner Source 72	E
	Solids Feed Rate per Batch		kg/batch		14,400	14,400	14,400	14,400	6	Calculated			
	Design Solids Feed Rate per Batch		kg/batch		15,000	15,000	15,000	15,000	5	Vendor	71		0
	Feed % Solids (after dewatering)		% w/w		90	90	90	90	5	Vendor	71	At commencement of leach	0
	Gravity Gold Recovery	Nominal	% Au		14	14	14	14	1	Owner	16, 72	Relative to Feed Grade	K
		Design	% Au		20	20	20	20	1	Owner	16, 72	Relative to Feed Grade	A
	Gravity Gold Recovery to Concentrate per Day	Nominal	g Au		16,701	18,015	18,015	18,550	6	Calculated			
		Design	g Au		23,858	25,736	25,736	26,500	6	Calculated			
	Gold Feed Grade	Nominal	ppm Au		1,160	1,251	1,251	1,288	6	Calculated			
		Design	ppm Au		1,591	1,716	1,716	1,767	6	Calculated			
	Gravity Leach Recovery	Range	%		92 - 98	92 - 98	92 - 98	92 - 98	1	Owner	16		E
		Design	%		98	98	98	98	1	Owner	16, 72	Value used in SysCAD model	K
	Cycle Time		h		12	12	12	12	1	Owner	16		E
	Pregnant Solution Volume		m³		14.5	14.5	14.5	14.5	5	Vendor	71		0
	Pregnant Solution Grade	Nominal	ppm Au		1,129	1,218	1,218	1,254	6	Calculated			
		Design	ppm Au		1,613	1,739	1,739	1,791	6	Calculated			
		Average	ppm Ag		29.3	29.3	29.3	29.3	1	Owner	33	Plant assays from Client	K
		Average	ppm Cu		302.9	302.9	302.9	302.9	1	Owner	33	Plant assays from Client	K
	Scheme Water Required	per batch	m3		12	12	12	12	6	Calculated			
	Residue (Tailings) % Solids		%w/w		40.0	40.0	40.0	40.0	5	Vendor	71		0

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
6.2	**Gravity Electrowinning**												
	Type				Clean in place	Clean in place	Clean in place	Clean in place	1	Owner	16		E
	Number of Circuits		No.		1	1	1	1	1	Owner	16		A
	Number Cells per Circuit		No.		1	1	1	1	1	Owner	16		A
	Number of Acacia Batches per day		No.		2	2	2	2	1	Owner	70		0
	Electrowinning Time		h		10.0	10.0	10.0	10.0	5	Vendor	33, 72, 73	Two electrowinning runs as required per day as per Owner Source 70 & 72	H
	Barren Solution Assay		ppm Au		<5	<5	<5	<5	1	Owner	16, 70		0
	Electrowinning Cell	Model			1000x1000-22D split	1000x1000-22D split	1000x1000-22D split	1000x1000-22D split	5	Vendor	73	2 compartments	0
	Number of Cathodes				22	22	22	22	5	Vendor	73	11 per compartment	0
	Feed Flowrate	Total	m^3/hr		48	48	48	48	5	Vendor	73	24m3/hr to each compartment for a 48m3/hr total. Vendor advised this is dependent on silver and copper grades. This may need to be reduced for higher silver grades to reduce detachment.	0
	Cell Linear Velocity		m/min		0.4	0.4	0.4	0.4	5	Vendor	73	Vendor advised this is dependent on silver and copper grades	0
	Current Density		A/m^2		15-25	15-25	15-25	15-25	5	Vendor	73		0
	Rectifier	per compartment	A		2500	2500	2500	2500	5	Vendor	73	2 off, one per compartment	0
7.0	**FLOTATION**												
7.1	**General**												
	Throughput		t/h		3,146	2,962	3,146	3494	6	Calculated			
	Feed Particle Size p80		µm		180	180	180	220	1	Owner	26		H
		Design	µm		220	220	220	220	1	Owner	26		0
	Froth Factors												
	Rougher Flotation Concentrate				2.00	2.00	2.00	2.00	1	Owner	16		E
	Scavenger Flotation Concentrate				1.60	1.60	1.60	1.60	1	Owner	16		E
	Cleaner Flotation Concentrate				2.50	2.50	2.50	2.50	1	Owner	16		E
	Cleaner Scavenger Flotation Concentrate				2.00	2.00	2.00	2.00	1	Owner	16		E
	Flotation Cell Transfer Lines				1.20	1.20	1.20	1.20	1	Owner	16		E
	Launders				-	-	-	-	1	Owner	16	Open launders - design to ensure no overflow	E
	Spray Water Line Pressure (at vendor battery limit)		kPa		82.74	82.74	82.74	82.74	5	Vendor	60		K
7.2	**Rougher Flotation**												
	Feed Streams				Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	1	Owner	43	Owner Supplied	A
	Solids Feed Rate	New Feed	t/h		3,146	2,962	3,146	3,494	6	Calculated			
		Total	t/h		3,161	2,977	3,160	3,512	6	Calculated		Includes sample return	
	Volumetric Feed Rate	Total	m^3/h		6,729	6,336	6,726	6,035	6	Calculated			
	Feed Solids Concentration	Total	% w/w		34.4	34.4	34.4	40.5	6	Calculated			
	Particle Size p80	Design	µm		220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative)	0
	Mass Pull		%		4.4	4.4	4.4	4.4	6	Calculated			
	Concentrate 1 Upgrade Ratio				15	15	15	15	1	Owner	16		K
	Concentrate 1 Grade		% S % Au in flotation feed		15.3	22.4	21.0	15.3	6	Calculated			
	Concentrate 1 Stage Recovery		% of S in flotation feed		65	65	65	65	1	Owner	23		G
	Concentrate 1 Stage Recovery				65	65	65	65	1	Owner	23		G
	Concentrate 1 Production		t/h		138	130	138	153	6	Calculated			
	Concentrate 1 Solids Concentration		% w/w		25	25	25	25	1	Owner	16		A
	Concentrate 1 Slurry Volume - Excl Spray Water		m^3/h		428	400	426	475	6	Calculated			
	Concentrate 1 SG				3.32	3.54	3.50	3.32	2	EPC	32		I
	Concentrate 1 Launder Spray Water	Operating Range	m^3/h		6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60		K
		Design	m^3/h		50	50	50	50	5	Vendor	60		K
	Mass Pull		%		2.4	2.4	2.4	2.4	6	Calculated			
	Concentrate 2 & 3 Upgrade Ratio				7	7	7	7	1	Owner	16		E
	Concentrate 2 & 3 Grade		% S % Au in flotation feed		7.1	10.4	9.8	7.1	6	Calculated			
	Concentrate 2 & 3 Stage Recovery		% of S in flotation feed		17	17	17	17	1	Owner	23		G
	Concentrate 2 & 3 Stage Recovery				17	17	17	17	1	Owner	23		G
	Concentrate 2 & 3 Production		t/h		77	73	77	86	6	Calculated			
	Concentrate 2 & 3 Solids Concentration		% w/w		25	25	25	25	1	Owner	16		A
	Concentrate 2 & 3 Slurry Volume - Excl Spray Water		m^3/h		242	227	241	268	6	Calculated			
	Concentrate 2 & 3 SG				3.06	3.17	3.15	3.06	2	EPC	32		I
	Concentrate 2 Launder Spray Water	Operating Range	m^3/h		6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60		K
	Concentrate 3 Launder Spray Water	Operating Range	m^3/h		6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60		K
	Concentrate 2 & 3 Launder Spray Water	Design (Combined)	m^3/h		50	50	50	50	5	Vendor	60		K
	Rougher Tail Solids Rate		t/h		2,946	2,774	2,946	3,273	6	Calculated			
	Rougher Tail Slurry Volumetric Flow		m^3/h		6,060	5,709	6,059	5,292	6	Calculated			
	Cell Type				Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Owner	16		A
	Aspiration				Forced	Forced	Forced	Forced	1	Owner	16		A
	Number Cells Installed				3	3	3	3	1	Owner	41		A
	Nominal Cell Volume		m^3		630	630	630	630	1	Owner	16		A
	Total Cell Volume		m^3		1,890	1,890	1,890	1,890	6	Calculated			
	Cell Air Hold-Up		%		15	15	15	15	1	Owner	41		A
	Plant Residence Time		min		15.1	16.0	15.1	17.0	6	Calculated		Rougher residence time existing plant = 10-12 mins (source 23). Residence time has been calculated using the average of the feed and tailings flowrates.	
	Lip Loading - Design		t/h/m		1.5	1.5	1.5	1.5	1	Owner	16		A
	Froth Carry Rate - Design		$t/m^2/h$		1.5	1.5	1.5	1.5	1	Owner	16		A
7.3	**Scavenger Flotation**												
	Feed Streams				Rougher Tailings	Rougher Tailings	Rougher Tailings	Rougher Tailings	1	Owner	43	Owner Supplied	A
	Solids Feed Rate		t/h		2,946	2,774	2,946	3,273	6	Calculated			
	Volumetric Feed Rate		m^3/h		6,060	5,709	6,060	5,292	6	Calculated			
	Feed Solids Concentration		% w/w		35.3	35.3	35.3	42.4	6	Calculated			
	Particle Size p80	Design	µm		220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative)	0
	Mass Pull (Scavenger Concentrate)		%		1.1	1.1	1.1	1.1	6	Calculated			
	Scavenger Concentrate Upgrade Ratio				5	5	5	5	1	Owner	16		E
	Scavenger Concentrate Grade		% S % Au in flotation feed		5.1	7.5	7.0	5.1	6	Calculated			
	Scavenger Concentrate Stage Recovery		% of S in flotation feed		5	5	5	5	1	Owner	23		G
	Scavenger Concentrate Stage Recovery				5	5	5	5	1	Owner	23		G
	Scavenger Concentrate Production		t/h		32	30	32	35	6	Calculated			
	Scavenger Concentrate Solids Concentration		% w/w		25	25	25	25	2	EPC	32		G
	Scavenger Concentrate Slurry Volume - Excl Spray Water		m^3/h		99	93	99	110	6	Calculated			
	Scavenger Concentrate SG				3.00	3.07	3.06	3.00	2	EPC	32		I
	Scavenger Concentrate Solids Concentration - Diluted		% w/w		20	20	20	20	2	EPC	32		H
	Scavenger Concentrate Launder Spray Water	Actual (Combined)	m^3/h		29	28	29	33	6	Calculated			
		Operating Range	m^3/h/cell		6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60		K
		Design	m3/h/cell		25.00	25.00	25.00	25.00	5	Vendor	60		K
	Scavenger Tail Solids Rate		t/h		2,914	2,745	2,914	3,238	6	Calculated			K
	Scavenger Tail Slurry Volumetric Flow		m^3/h		5,961	5,616	5,961	5,182	6	Calculated			K
	Cell Type				Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Owner	16		A
	Aspiration				Forced	Forced	Forced	Forced	1	Owner	16		A
	Number Cells Installed				4	4	4	4	1	Owner	41		A
	Nominal Cell Volume		m^3		630	630	630	630	1	Owner	16		A
	Total Cell Volume		m^3		2,520	2,520	2,520	2,520	6	Calculated			
	Cell Air Hold-Up		%		15	15	15	15	1	Owner	41		A
	Plant Residence Time		min		21.4	22.7	21.4	24.5	6	Calculated		Scavenger residence time existing plant = 29 - 32 mins (source 23). Residence time has been calculated using the average of the feed and tailings flowrates.	
	Lip Loading - Design		t/h/m		1.5	1.5	1.5	1.5	1	Owner	16		A
	Froth Carry Rate - Design		$t/m^2/h$		1.5	1.5	1.5	1.5	1	Owner	16		A
	Combined Rougher and Scavenger Residence Time				36.5	38.7	36.5	41.6	6	Calculated			

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
7.4	**Cleaner Flotation**											Owner Supplied	
	Feed Streams				Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	1	Owner	43		A
	Solids Feed Rate	New Feed Only	t/h		253	244	258	282	6	Calculated		New feed only	
		Total	t/h		465	424	439	478	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA	
	Volumetric Feed Rate to Cell - Target	Total	m³/h		1998	1998	1998	1998	5	Vendor	27	Average of four modelled cases	1
	Volumetric Feed Rate to Cell - Actual	Total	m³/h		1,998	1,998	1,998	1,998	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA	
	Feed Solids Concentration - Actual	Total	% w/w		19.0	17.5	18.1	19.5	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA	
	Particle Size p80	Design	µm		220	220	220	220	6	Calculated		Feed, concentrate and tailings p80 assumed equal (conservative). Max of calculated p80	
		Calculated	µm		180	180	180	220	6	Calculated			
	Mass Pull		%		30.8	43.9	41.5	30.7	6	Calculated		Wrt to new feed	
	Concentrate Grade		% S		35	35	35	35	1	Owner	26		G
	Concentrate Stage Recovery		% S		93.3	93.5	93.3	93.1	6	Calculated		Wrt to new feed	
	Concentrate Recovery	Overall	% of S in flotation feed		85.1	85.1	85.1	85.1	1	Owner	23		G
	Concentrate Recovery	Overall	% Au in flotation feed		71.8	71.8	71.8	71.8	1	Owner	23		G
	Concentrate Production		t/h		78	107	107	86	6	Calculated			
	Concentrate Solids Concentration		% w/w		25	25	25	25	5	Vendor	27	Undiluted cleaner flotation concentrate	G
	Concentrate Slurry Volume - Excl Spray Water		m³/h		238	327	327	264	6	Calculated			
	Concentrate SG				3.93	3.93	3.93	3.93	2	EPC	32		I
	Concentrate Wash Water Ratio (of Con m3/hr)				1.0	1.0	1.0	1.0	5	Vendor	27		I
	Cleaner Tail Solids Rate		t/h		183	145	159	203	6	Calculated		To Cleaner-Scavenger Flotation Cell Feed Hopper	
	Cleaner Tail Slurry Volumetric Flow		m³/h		946	913	960	1,039	6	Calculated		To Cleaner-Scavenger Flotation Cell Feed Hopper	
	Cleaner Tails Recycle		%		52.6	54.3	52.0	48.0	6	Calculated		The % of total tailings recycled back to feed hopper.	
	Cleaner Tail Solids Rate		t/h		204	172	172	188	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cleaner Tail Slurry Volumetric Flow		m³/h		1,052	1,085	1,038	959	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cell Type				Jameson	Jameson	Jameson	Jameson	5	Vendor	18		E
	Model				B6500/24	B6500/24	B6500/24	B6500/24	5	Vendor	27		G
	Aspiration Type				Natural	Natural	Natural	Natural	1	Owner	16		E
	Number Cells Installed				1	1	1	1	5	Vendor	18		E
	Number of Downcomers				24	24	24	24	5	Vendor	27		0
	Froth Area		m²		22.2	22.2	22.2	22.2	5	Vendor	27		G
	Froth Carry Rate	Nom	t/m²/h		3.50	4.81	3.50	3.51	5	Vendor	27		1
		Max	t/m²/h		6.5	6.5	6.5	6.5	5	Vendor	27		G
		Calculated	t/m²/h		3.51	4.82	4.82	3.89	6	Calculated			
7.5	**Cleaner Scavenger Flotation**											Owner Supplied	
	Feed Streams				Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	1	Owner	66		0
	Solids Feed Rate	New Feed Only	t/h		183	145	159	203	6	Calculated		New feed only	
		Total	t/h		379	300	317	385	6	Calculated		Total feed to Jameson cell including tails recycle	
	Volumetric Feed Rate to Cell - Target	Total	m³/h		2048	2048	2048	2048	5	Vendor	27	Average of four modelled cases	1
	Volumetric Feed Rate to Cell - Actual	Total	m³/h		2,048	2,048	2,048	2,048	6	Calculated		Total feed to Jameson cell including tails recycle	
	Feed Solids Concentration	Total	% w/w		15.6	12.6	13.2	15.8	6	Calculated		Total feed to Jameson cell including tails recycle	
	Particle Size p80	Design	µm		220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative). Max of calculated p80	0
	Mass Pull		%		7.6	13.2	12.0	7.6	6	Calculated		Wrt to new feed	
	Concentrate Grade		% S		13.10	13.10	13.10	13.10	5	Vendor	19	Design Grade sulphur grade based on EPC circuit modelling	E
	Concentrate Stage Recovery		% S		49.6	53.6	53.7	50.9	6	Calculated			
	Concentrate Recovery		% of Au in Flotation Feed		5.7	5.7	5.7	5.7	3	Assumed			K
	Concentrate Recovery		% of S in Flotation Feed		5.7	5.7	5.7	5.7	2	EPC	32		H
	Concentrate Production		t/h		14	19	19	15	6	Calculated			
	Concentrate Solids Concentration		% w/w		25	25	25	25	2	EPC	32		I
	Concentrate Slurry Volume - Excl Spray Water		m³/h		43	60	60	48	6	Calculated			
	Concentrate Solid SG				3.25	3.25	3.25	3.25	2	EPC	32		I
	Concentrate Wash Water Ratio (of Con m3/hr)				1	1	1	1	5	Vendor	27		I
	Cleaner Scavenger Tail Solids Rate		t/h		169	126	140	188	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper	
	Cleaner Scavenger Tail Slurry Volumetric Flow		m³/h		950	916	963	1,043	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper	
	Cleaner Scavenger Tails Recycle		%		53.6	55.3	53.0	49.1	6	Calculated		The % of total tailings recycled back to feed hopper.	
	Cleaner Scavenger Tail Solids Rate		t/h		196	155	158	181	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cleaner Scavenger Tail Slurry Volumetric Flow		m³/h		1,098	1,132	1,085	1,005	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cell Type				Jameson	Jameson	Jameson	Jameson	5	Vendor	19		E
	Model				B6500/24	B6500/24	B6500/24	B6500/24	5	Vendor	27		H
	Aspiration Type				Natural	Natural	Natural	Natural	1	Owner	16		E
	Number Cells Installed				1	1	1	1	5	Vendor	19		E
	Number of Downcomers				24	24	24	24	5	Vendor	27		0
	Froth Area		m²		22.2	22.2	22.2	22.2	5	Vendor	27		G
	Froth Carry Rate	Nom	t/m²/h		0.63	0.85	0.63	0.63	5	Vendor	27		1
		Max	t/m²/h		6.5	6.5	6.5	6.5	5	Vendor	27		G
		Calculated	t/m²/h		0.63	0.86	0.86	0.70	6	Calculated			

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
8.0	**CONCENTRATE HANDLING**												
8.1	**Concentrate Thickener (Brownfields)**				Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)				Existing 30-TH-61	
	Feed Stream				Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	1	Owner	16		E
	Solids Feed Rate		t/h		75	104	104	84	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant		m³/h		230	319	319	256	6	Calculated			
	Feed Solids Concentration		% w/w		24.9	24.9	24.9	24.9	6	Calculated			
	Underflow Solids Concentration		% w/w		60	60	60	60	1	Owner	16		E
	Testwork Flux Rate		$t/h/m^2$		0.25	0.25	0.25	0.25	1	Owner	23		G
	Calculated Flux Rate		$t/h/m^2$		0.13	0.18	0.18	0.15	6	Calculated			
	Thickener Area Calculated		m^2		573	573	573	573	6	Calculated			
	Diameter	Existing	m		27	27	27	27	1	Owner	16		A
	Structural Design SG				3.8	3.8	3.8	3.8	1	Owner	16		E
	Concentrate Thickener Sprays - Mill Circuit Water		m³/h		68.40	68.40	68.40	68.40	5	Vendor	113		1
8.2a	**Filtered Feed Tank A**				Existing	Existing	Existing	Existing					
	Feed Streams				Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	1	Owner	43		E
	Solids Feed Rate		t/h		75	104	104	84	6	Calculated			
	Volumetric Feed Rate		m³/h		66	92	92	74	6	Calculated			
	Feed Solids Concentration		% w/w		59.6	59.7	59.7	59.6	6	Calculated			
	Number of Tanks				1	1	1	1	1	Owner	16		E
	Residence Time		h		10.6	7.6	7.6	9.5	6	Calculated			
	Tank Capacity		m³		700	700	700	700	1	Owner	16		E
8.2b	**Filtered Feed Tank B**				Existing	Existing	Existing	Existing					
	Feed Streams				Concentrate Deslime Cyclone Underflow (Stage 1 Only) UFG Feed Preparation Cyclone Underflow (Stage 1 Only)	Concentrate Deslime Cyclone Underflow (Stage 1 Only) UFG Feed Preparation Cyclone Underflow (Stage 1 Only)	Concentrate Deslime Cyclone Underflow (Stage 1 Only) UFG Feed Preparation Cyclone Underflow (Stage 1 Only)	Concentrate Deslime Cyclone Underflow (Stage 1 Only) UFG Feed Preparation Cyclone Underflow (Stage 1 Only)	1	Owner	16		E
	Solids Feed Rate		t/h		129	153	153	149	6	Calculated			
	Volumetric Feed Rate		m³/h		98	129	129	89	6	Calculated			
	Feed Solids Concentration		% w/w		64.7	61.4	61.4	73.0	6	Calculated			
	Number of Tanks				1	1	1	1	1	Owner	16		E
	Residence Time		h		7.2	5.4	5.4	7.9	6	Calculated			
	Tank Capacity		m³		700	700	700	700	1	Owner	16		E
8.3	**Concentrate Deslime Cyclones (Brownfields, Stage 1 Only)**				Existing	Existing	Existing	Existing					
	Feed Stream				Filtered Feed Tank A	Filtered Feed Tank A	Filtered Feed Tank A	Filtered Feed Tank A	1	Owner	43		E
	Type				Cyclone	Cyclone	Cyclone	Cyclone	1	Owner	16		E
	Cyclone Size		mm		100	100	100	100	1	Owner	16		E
	Cyclone Model				Concord Kamlock	Concord Kamlock	Concord Kamlock	Concord Kamlock	1	Owner	16		E
	Number of Cyclone Clusters				1	1	1	1	1	Owner			E
	Number Installed Cyclones		No.		36	36	36	36	1	Owner	16		E
	Cyclone Solids Feed Rate		t/h		45	74	74	38	6	Calculated			
	Cyclone Feed Flow		m³/h		179	252	252	150	6	Calculated			
	Cyclone Feed Solids Concentration	Range	% w/w		15 - 35	15 - 35	15 - 35	15 - 35	1	Owner	16	In-line dilution of Concentrate Deslime Cyclone feed pumps required	K
		Nominal	% w/w		20	20	20	20	1	Owner	16	In-line dilution of Concentrate Deslime Cyclone feed pumps required	E
	Underflow Solids Concentration	Range	% w/w		39 - 50	39 - 50	39 - 50	39 - 50	1	Owner	16		
		Nominal	% w/w		58.0	58.0	58.0	58.2	6	Calculated	58		E
	Underflow Solids		t/h		38	63	63	32	6	Calculated			
	Underflow Mass Split		%		85	85	85	85	1	Owner	16		E
	Au % Recovery to Deslime Cyclone Underflow	wrt to Deslime Cyclone Feed	%		75	75	75	75	1	Owner	44	Range 71 - 78%	K
	S % Recovery to Deslime Cyclone Underflow	wrt to Deslime Cyclone Feed	%		82	82	82	82	1	Owner	44	Range 79 - 86%	K
	Overflow Solids Concentration		% w/w		4.2	5.2	5.2	4.2	6	Calculated			
	Overflow Solids		t/h		7	11	11	6	6	Calculated			
	Operating Pressure		kPa		200 - 240	200 - 240	200 - 240	200 - 240	1	Owner	16		E
	Product Size P_{80}		µm		11.1	11.1	11.1	11.1	1	Owner	16		E
8.4	**Concentrate Filters (Brownfields)**				Existing	Existing	Existing	Existing					
	Feed Stream				Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	1	Owner	43		E
	Filter Type				Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter					E
	Number of Filters				2	2	2	2	1	Owner	16		E
	Solids Feed Rate		t/h		32	59	59	22	6	Calculated			
	Volumetric Feed Rate		m³/h		36	69	69	24	6	Calculated			
	Feed Solids Concentration	Target	% w/w		60.0	60.0	60.0	60.0	1	Owner	16		E
		Actual	% w/w		51.1	50.3	50.3	52.8	6	Calculated			
	Target Moisture Content		< % w/w		10.0	10.0	10.0	10.0	1	Owner	16		E
	Filter Area	per filter	m^2		55.0	55.0	55.0	55.0	1	Owner	16		E
	Specific Filtration Rate	Nominal	$m^3/h/m^2$		0.14	0.14	0.14	0.14	1	Owner	16		E
		Design	$m^3/h/m^2$		0.31	0.31	0.31	0.31	1	Owner	16		E
8.5	**Concentrate Stockpile**												
	Feed Stream				Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake	1	Owner	43		E
	No. of Stockpiles				1	1	1	1	1	Owner	16		E
	Concentrate Bulk Density		t/m^3		2.2	2.2	2.2	2.2	1	Owner	16		E
	Angle of Repose	Nominal	Degrees		40.0	40.0	40.0	40.0	1	Owner	16		E
	Storage		days		2	2	2	2	1	Owner	16		E
	Capacity		m^3		1,139	1,139	1,139	1,139	1	Owner	16		E
	TML for Bulk Shipment		% w/w		6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	1	Owner	23		G
8.6	**Slimes Flotation (Brownfields, Stage 1 Only)**				Existing	Existing	Existing	Existing					
	Feed Streams				Concentrate Deslime Cyclone Overflow	Concentrate Deslime Cyclone Overflow	Concentrate Deslime Cyclone Overflow	Concentrate Deslime Cyclone Overflow	1	Owner	16		E
	Solids Feed Rate		t/h		7	11	11	6	6	Calculated			
	Volumetric Feed Rate		m³/h		144	194	194	120	6	Calculated			
	Target Feed Solids Concentration		% w/w		9.9	9.9	9.9	9.9	1	Owner	16		E
	Mass Pull		%		57	57	57	57	6	Calculated		Output of SysCAD to achieve the Au and S Recoveries	
	Concentrate Production		t/h		4	6	6	3	6	Calculated			
	Concentrate Solids Concentration		% w/w		20	20	20	20	1	Owner	16		E
	Concentrate SG		t/m^3		3.1	3.1	3.1	3.1	1	Owner	16, 44	Owner provided range 2.8 - 3.2	K
	Concentrate Spray Water		m³/h		0	0	0	0	1	Owner	44	No spray water used	K
	Au % Recovery to Slimes Flotation Concentrate	wrt to Slimes Flotation Feed	%		80	80	80	80	1	Owner	44	Range 76 - 86%	K
	S % Recovery to Slimes Flotation Concentrate	wrt to Slimes Flotation Feed	%		72	72	72	72	1	Owner	44	Range 65 - 78%	K
	Tail Rate		t/h		3	5	5	2	6	Calculated			
	Tail Slurry Volume		m³/h		129	169	169	108	6	Calculated			
	Cell Type				Conventional	Conventional	Conventional	Conventional	1	Owner	16		E
	Aspiration				Forced	Forced	Forced	Forced	1	Owner	16		E
	Number Cells Installed				4	4	4	4	1	Owner	16		E
	Nominal Cell Volume		m³		8.5	8.5	8.5	8.5	1	Owner	16		E
	Total Cell Volume		m³		34	34	34	34	6	Calculated			
	Cell Air Hold-Up		%		15	15	15	15	1	Owner	16		E
9.0	**CONCENTRATE ULTRAFINE GRINDING**												
9.1	**UFG Feed Preparation Cyclones (Brownfields, Stage 1 Only)**				Existing	Existing	Existing	Existing					
	Cyclone Diameter		mm		150	150	150	150	1	Owner	16	These are solely used ahead of UFG Mill 1 (Existing M3000)	E
	Number Installed Cyclones		No.		9	9	9	9	1	Owner	90		1
	Number Operating Cyclones		No.		4	4	4	4	1	Owner	90		1
	Cyclone Solids Feed Rate		t/h		100	100	100	130	6	Calculated			
	Cyclone Feed Flow		m³/h		76	84	84	78	6	Calculated			
	Cyclone Feed Solids Concentration		% w/w		64.7	61.4	61.4	73.0	6	Calculated			
	Underflow Solids Concentration		% w/w		74	74	74	74	1	Owner	16		E
	Underflow Solids		t/h		90	90	90	117	6	Calculated			
	Overflow Solids Concentration		% w/w		45	45	45	45	1	Owner	16		E
	Overflow Solids		t/h		10	10	10	13	6	Calculated			
	Overflow Sulphur Grade		%		30.76	27.40	27.40	30.76	1	Owner	44	Based on existing plant data. Nominal values for mass balance purposes. 80th percentile (50th 27.4%). Raw Assay Data YTD	K
	Overflow Gold Grade		g/t		21.20	19.50	19.50	21.20	1	Owner	44	Based on existing plant data. Nominal values for mass balance purposes. 80th percentile (50th 19.5g/t). Raw Assay Data YTD	K
	Cyclone UF/Cyclone OF		%		904	904	904	904	6	Calculated	58		E
	Operating Pressure		kPa		100	100	100	100	1	Owner	16		E
9.2	**Concentrate UFG Mill 1 (Brownfields)**				Existing	Existing	Existing	Existing				M3000	
	Circuit Feed Streams				Concentrate Filter Feed Tank B	Concentrate Filter Feed Tank B	Concentrate Filter Feed Tank B	Concentrate Filter Feed Tank B	1	Owner	16, 80		E
	Feed Rate - Fresh Feed	P_{80} 9 µm	t/h		10.0	10.0	10.0	10.0	1	Owner	69		0
		Design	t/h		13.0	13.0	13.0	13.0	1	Owner	16		E
	Feed Solids Concentration		% w/w		45	45	45	45	1	Owner	4		0
	Feed F_{80}	Nominal	µm		144	144	144	144	1	Owner	16		E
	Specific Energy	P_{80} 9 µm	kWh/t		116.5	116.5	116.5	116.5	1	Owner	16		A
	Type				IsaMill	IsaMill	IsaMill	IsaMill	1	Owner	16		A
	Model				M3000	M3000	M3000	M3000	1	Owner	16		A
	Number				1	1	1	1	1	Owner	16		A
	Configuration				Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Owner	16		A
	Mill Power Draw	Nominal	kW		860	860	860	860	1	Owner	16		A
	Installed Motor Power		kW		1,120	1,120	1,120	1,120	1	Owner	16		A
	UFG Mill Recycle Proportion		%		0	0	0	0	1	Owner	4		A
	UFG Circuit Product Size P_{80}	Nominal	µm		9.0	9.0	9.0	9.0	1	Owner	16		A
	UFG Circuit Product Size P_{80} - Range		µm		9 - 12	9 - 12	9 - 12	9 - 12	1	Owner	16		E
	Gland Seal Water		m³/h		1.2	1.2	1.2	1.2	1	Owner	2		A
	Flushing Gland Seal Water		m³/h		5.4	5.4	5.4	5.4	1	Owner	2		A

AREA	PARAMETER		UNIT	FIM	DATA 1 NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	DATA 2 NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	DATA 3 NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	DATA 4 NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um	REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
9.3	**UFG Mill 1 Trash Screens (Brownfields)**				Existing	Existing	Existing	Existing					
	Feed Stream				UFG Mill 1 Discharge	UFG Mill 1 Discharge	UFG Mill 1 Discharge	UFG Mill 1 Discharge	1	Owner	43		E
	Type				Vibrating Screen	Vibrating Screen	Vibrating Screen	Vibrating Screen	1	Owner	16		E
	Number of Screens per UFG Circuit		No.		1	1	1	1	1	Owner	16		E
	Screen Solids Feed Rate per Screen	P_{80} 9 µm	t/h		60.0	60.0	60.0	78.0	6	Calculated			
	Screen Feed Density	P_{80} 9 µm	% w/w		42.7	42.7	42.7	43.2	6	Calculated			
	Screen Feed Flowrate per Screen	P_{80} 9 µm	m^3/h		90.9	91.5	91.5	116.2	6	Calculated			
	Screen Size	W x L	m		1.2 x 2.4	1.2 x 2.4	1.2 x 2.4	1.2 x 2.4	1	Owner	16		E
	Screen Area per Screen		m^2		2.9	2.9	2.9	2.9	6	Calculated			
	Specific Screen Flux Rate		$m^3/m^2/h$		31.6	31.8	31.8	40.3	6	Calculated			
	Screen Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		E
	Screen Material				Wire Mesh	Wire Mesh	Wire Mesh	Wire Mesh	1	Owner	23		G
	Screen Spray Water		m³/h		0	0	0	0	1	Owner	44		0
9.4	**UFG Mill 1 Classification (Brownfields)**				Existing	Existing	Existing	Existing					
	Cyclone Model				100CVX	100CVX	100CVX	100CVX	1	Owner	16		E
	Number of Cyclone Clusters per UFG Circuit				1	1	1	1	1	Owner	16		E
	Number Installed Cyclones per Cluster		No.		30	30	30	30	1	Owner	16		E
	Cyclone Solids Feed Rate - per UFG circuit	P_{80} 9 µm	t/h		60.0	60.0	60.0	78.0	6	Calculated			
	Cyclone Feed Flow - per UFG circuit	P_{80} 9 µm	m³/h		99.9	100.4	100.4	129.8	6	Calculated			
	Cyclone Feed Solids Concentration		% w/w		40	40	40	40	1	Owner	4		0
	Underflow Solids Concentration		% w/w		60	60	60	60	1	Owner	5		A
	Underflow Solids - per UFG circuit	P_{80} 9 µm	t/h		50.0	50.0	50.0	65.0	6	Calculated			
	Overflow Solids Concentration	P_{80} 9 µm	% w/w		15.0	15.0	15.0	15.0	6	Calculated			
	Overflow Solids - per UFG circuit	P_{80} 9 µm	t/h		10.0	10.0	10.0	13.0	6	Calculated			
	Cyclone Circulating Load		%		500	500	500	500	1	Owner	16		E
	Operating Pressure		kPa		300	300	300	300	1	Owner	16		C
	Product Size P_{80}		µm		9	9	9	9	1	Owner	5		A
9.5	**New Concentrate UFG Mill 2**				Greenfields	Greenfields	Greenfields	Greenfields				M15,000	G
	Circuit Feed Streams				Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	1	Owner	80	Runs in parallel with the existing M3000	G
	Nominal Feed Rate - Fresh Feed	P_{80} 9 µm	t/h		30.0	30.0	30.0	30.0	1	Owner	24		G
	Maximum Feed Rate - Fresh Feed	P_{80} 12 µm	t/h		46.0	46.0	46.0	46.0	1	Owner	24		G
	Total Feed Solids Concentration		% w/w		47.5	47.5	47.5	47.7	5	Vendor	65	Includes Recycle Portion	0
	Feed F_{80}	Nominal	µm		144	144	144	144	1	Owner	16		E
	Specific Energy	P_{80} 9 µm	kWh/t		116.5	116.5	116.5	116.5	1	Owner	16		A
	Type				IsaMill	IsaMill	IsaMill	IsaMill	1	Owner	23		G
	Model				M15,000	M15,000	M15,000	M15,000	1	Owner	23		G
	Number				1	1	1	1	1	Owner	23		G
	Configuration				Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Owner	23		G
	Mill Power Draw	Nominal	kW		3,500	3,500	3,500	3,500	1	Owner	5		A
	Installed Motor Power		kW		3,800	3,800	3,800	3,800	1	Owner	23		G
	UFG Mill Recycle Proportion		%		20	20	20	5	5	Vendor	65	% of New (Fresh) UFG Mill 2 Feed	0
	UFG Circuit Product Size P_{80}	Nominal	µm		9.0	9.0	9.0	9.0	1	Owner	16		A
	UFG Circuit Product Size P_{80} - Range		µm		9 - 12	9 - 12	9 - 12	9 - 12	1	Owner	16		E
	Gland Seal Water		m³/h		2.0	2.0	2.0	2.0	5	Vendor	65	Scheme Water	0
	Flushing Gland Seal Water		m³/h		5.4	5.4	5.4	5.4	5	Vendor	65	Scheme Water	0
9.6	**UFG Mill 2 Trash Screens**				Greenfields	Greenfields	Greenfields	Greenfields					
	Feed Stream				UFG Mill 2 Fresh Feed	UFG Mill 2 Fresh Feed	UFG Mill 2 Fresh Feed	UFG Mill 2 Fresh Feed	1	Owner	23		G
	Type				Inclined Static Screen	Inclined Static Screen	Inclined Static Screen	Inclined Static Screen	1	Owner	74		0
	Number of Screens per UFG Circuit		No.		1	1	1	1	1	Owner	23		G
	Screen Solids Feed Rate per Screen	P_{80} 9 µm	t/h		30.0	30.0	30.0	46.0	6	Calculated			
	Screen Feed Density	P_{80} 9 µm	% w/w		58.5	58.6	58.6	58.9	6	Calculated			
	Screen Feed Flowrate per Screen	P_{80} 9 µm	m^3/h		27.4	27.3	27.3	41.5	6	Calculated			
	Screen Size	L x W	m		1.240 x 1.6	1.240 x 1.6	1.240 x 1.6	1.240 x 1.6	5	Vendor	77		0
	Effective Screening Area per Screen		m^2		2.84	2.84	2.84	2.84	5	Vendor	77		0
	Specific Screen Flux Rate		$m^3/m^2/h$		12.80	12.77	12.77	17.76	6	Calculated		Includes wash water	0
	Screen Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		E
	Screen Material				Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	5	Vendor	77		0
	Screen Spray Water		m³/h		9.0	9.0	9.0	9.0	5	Vendor	77		1
9.7	**UFG Mill 2 Classification**				Greenfields	Greenfields	Greenfields	Greenfields					
	Cyclone Model				100CVX	100CVX	100CVX	100CVX	1	Owner	16		E
	Number of Cyclone Clusters per UFG Circuit				1	1	1	1	5	Vendor	54		G
	Number Installed Cyclones per Cluster		No.		24	24	24	24	5	Vendor	54		0
	Number of Cyclone Distributor Outlets Available		No.		24	24	24	24	5	Vendor	54		A
	Spare Outlets Available		No.		0	0	0	0	6	Calculated			
	Number of Operating Cyclones	P_{80} 9 µm	No.		14	14	14	19	5	Vendor	54		0
	Number of Standby Cyclones	P_{80} 9 µm	No.		10	10	10	5	6	Calculated			
	Cyclone Solids Feed Rate - per UFG circuit	P_{80} 9 µm	t/h		118.0	118.0	118.0	161.0	6	Calculated			
	Cyclone Feed Flow - per UFG circuit	P_{80} 9 µm	m³/h		251.7	251.7	251.6	299.8	6	Calculated			
	Cyclone Feed Solids Concentration		% w/w		32.8	32.8	32.8	32.9	5	Vendor	65		0
	Underflow Solids Concentration		% w/w		48.0	48.0	48.0	48.0	5	Vendor	65		0
	Underflow Solids - per UFG circuit	P_{80} 9 µm	t/h		88.0	88.0	88.0	115.0	6	Calculated			
	Overflow Solids Concentration	P_{80} 9 µm	% w/w		17.4	17.5	17.5	23.1	6	Calculated			
	Overflow Solids - per UFG circuit	P_{80} 9 µm	t/h		30.0	30.0	30.0	46.0	6	Calculated			
	Cyclone Circulating Load		%		293	293	293	250	5	Vendor	54		0
	Operating Pressure		kPa		247	247	247	248	5	Vendor	54		0
	Product Size P_{80}		µm		9	9	9	9	1	Owner	5		A
9.8	FIM **Concentrate Thickener (Brownfields, Stage 1 Only)**				Existing	Existing	Existing	Existing					
	Feed Streams				Slimes Flotation Tails UFG Circuit 1 Cyclone O/F UFG Circuit 2 Cyclone O/F	Slimes Flotation Tails UFG Circuit 1 Cyclone O/F UFG Circuit 2 Cyclone O/F	Slimes Flotation Tails UFG Circuit 1 Cyclone O/F UFG Circuit 2 Cyclone O/F	Slimes Flotation Tails UFG Circuit 1 Cyclone O/F UFG Circuit 2 Cyclone O/F	1	Owner	24		G
	Solids Feed Rate		t/h		42.9	44.8	44.8	61.4	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant		m³/h		351.4	414.9	414.9	351.8	6	Calculated			
	Feed Solids Concentration		% w/w		10.5	9.4	9.4	14.6	6	Calculated			
	Underflow Solids Concentration		% w/w		50	50	50	50	1	Owner	16		A
	Calculated Flux Rate		$t/h/m^2$	14.78641084	0.17	0.18	0.18	0.24	6	Calculated			
	Testwork Flux Rate		$t/h/m^2$	171.717841	0.25	0.25	0.25	0.25	1	Owner	23		G
	Thickener Area Calculated		m^2	18	254	254	254	254	6	Calculated			
	Diameter	Existing	m		18	18	18	18	1	Owner	16		A
	Thickener Spray Water		m³/h		0	0	0	0	1	Owner	44		0
10.0	**CIL - FLOTATION CONCENTRATE**												
10.1	**General Circuit (Brownfields)**				Existing CIL 1	Existing CIL 1	Existing CIL 1	Existing CIL 1					
	Solids Feed Rate	Fresh	t/h		43	45	45	61	6	Calculated		Based on P80 9 µm	
	Target Feed Pulp Density		% w/w		50	50	50	50	1	Owner	16		A
	Actual Feed Pulp Density		% w/w		37.4	37.6	37.6	38.9	6	Calculated			
	Slurry Feed Flow Rate	Fresh	m³/h		51	32	15	35	6	Calculated		Prior to reagent and eluate addition	
	Slurry Feed Flow Rate	Total	m³/h		82	85	85	111	6	Calculated		Inclusive of reagent and eluate addition	
	Leach Gold Extraction		%		84.0	84.0	84.0	84.0	1	Owner	44	Range 80 - 88%	K
	Adsorption Efficiency		%		99.0	99.0	99.0	99.0	1	Owner	44		K
10.2	**Leaching (Brownfields)**				Existing CIL 1	Existing CIL 1	Existing CIL 1	Existing CIL 1				1 train	
					FIM Concentrate Thickener Underflow	FIM Concentrate Thickener Underflow	FIM Concentrate Thickener Underflow	FIM Concentrate Thickener Underflow					
	Feed Streams								1	Owner	43		E
	Leach pH				11 - 12	11 - 12	11 - 12	11 - 12	1	Owner	16		E
	Number of Tanks				2	2	2	2	1	Owner	16		E
	Effective Tank Capacity	Per Tank	m^3		887	887	887	887	1	Owner	16		E
	Total Tank Capacity		m^3		1,774	1,774	1,774	1,774	6	Calculated			
	Residence Time		h		22	21	21	16	6	Calculated			
	Oxygen Addition Points				Leach Tanks and First Three Adsorption Tanks	Leach Tanks and First Three Adsorption Tanks	Leach Tanks and First Three Adsorption Tanks	Leach Tanks and First Three Adsorption Tanks	1	Owner	16		E
	Oxygen Addition Method				Side Lances in Leach Tanks and 3 Adsorption Tanks	Side Lances in Leach Tanks and 3 Adsorption Tanks	Side Lances in Leach Tanks and 3 Adsorption Tanks	Side Lances in Leach Tanks and 3 Adsorption Tanks	1	Owner	67		0
	Dissolved Oxygen Concentration	Target	ppm		10	10	10	10	1	Owner	16		E
10.3	**Adsorption (Brownfields)**				Existing CIL 1	Existing CIL 1	Existing CIL 1	Existing CIL 1					
	Number of Tanks				6	6	6	6	1	Owner	16		A
	Effective Tank Capacity	Per Tank	m^3		793	793	793	793	1	Owner	16		E
	Total Adsorption Tank Live Volume		m^3		4,758	4,758	4,758	4,758	6	Calculated			
	Residence Time		h		58.0	55.8	55.9	42.9	6	Calculated			
	Total Leach & Adsorption Time		h		80	77	77	59	6	Calculated			
	Carbon Concentration	per tank	g/L		20	20	20	20	1	Owner	16		E
	Carbon In Circuit	Total	t		95.2	95.2	95.2	95.2	6	Calculated			
	Design Carbon Loading		g/t		1,154	1,154	1,154	1,154	1	Owner	16		E
	Carbon Transfer Method				Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	1	Owner	16		E
	Carbon Transfer Period	per cycle	h		12	12	12	12	1	Owner	16		E
	Carbon and Pulp Transfer Rate		m³/h		60.0	60.0	60.0	60.0	1	Owner	44		0
	Loaded Carbon Advance Method				Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	1	Owner	16		E
	Loaded Carbon Pump & Pulp Forwarding Rate		m³/h		275	275	275	275	6	Calculated		Based on one 11 t strip per day	
	Loaded Carbon Transfer Time		h		2	2	2	2	1	Owner	16		E
10.4	**Loaded Carbon Recovery Screen (Brownfields)**				Existing CIL 1 Screen	Existing CIL 1 Screen	Existing CIL 1 Screen	Existing CIL 1 Screen					
	Feed Stream				First CIL 1 Adsorption Tank	First CIL 1 Adsorption Tank	First CIL 1 Adsorption Tank	First CIL 1 Adsorption Tank	1	Owner	16		
	Type				Vibrating Screen	Vibrating Screen	Vibrating Screen	Vibrating Screen	1	Owner	16		
	Number		No.		1	1	1	1	1	Owner	16	1 per CIL train	
	Feed Rate		m³/h		275	275	275	275	6	Calculated			
	Selected Size	Width	m		1.2	1.2	1.2	1.2	1	Owner	16		E
		Length	m		2.4	2.4	2.4	2.4	1	Owner	16		E
	Area		m^2		2.9	2.9	2.9	2.9	6	Calculated			
	Deck Type				Polyurethane	Polyurethane	Polyurethane	Polyurethane	1	Owner	16		E
	Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		E
	Spray Water Addition		m³/h		6	6	6	6	1	Owner	4		E
	Loaded Carbon Moisture		% w/w		40	40	40	40	1	Owner	4		E
	Loaded Carbon Transfer Moisture		% w/w		55	55	55	55	1	Owner	4	From Transfer Vessel	1

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
10.5	**Interstage Carbon Screens (Brownfields)**				Existing CIL 1	Existing CIL 1	Existing CIL 1	Existing CIL 1					
	Type				Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	1	Owner	16		A
	Model				MPSP 1200	MPSP 1200	MPSP 1200	MPSP 1200	1	Owner	16		E
	Number	per tank	No.		1	1	1	1	1	Owner	16		E
	Screen Feed Rate - per Screen	w/o carbon advance	m^3/h		82	85	85	111	6	Calculated			
		w/ carbon advance	m^3/h		142	145	145	171	6	Calculated			
	Area Selected	Selected	m^2		2.6	2.6	2.6	2.6	1	Owner	16		E
	Specific Screen Flux Rate	w/o carbon advance	$m^3/m^2/h$		31.6	32.8	32.8	42.6	6	Calculated			
		w/ carbon advance	$m^3/m^2/h$		54.7	55.8	55.8	65.7	6	Calculated			
	Screen Material				Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	1	Owner	16		E
	Screen Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		E
11.0	**FLOTATION TAILINGS THICKENING**												
11.1	**Flotation Tailings Thickener**											Owner Supplied	
	Feed Stream				Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	1	Owner	16		E
	Solids Feed Rate		t/h		3,071	2,858	3,042	3,411	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant		m^3/h		6,881	6,503	6,894	6,195	6	Calculated			
	Feed Solids Concentration		% w/w		33.0	32.6	32.7	38.9	6	Calculated			
	Underflow Solids Concentration	Nominal	% w/w		56.0	56.1	56.1	56.1	6	Calculated		Back-calculated to achieve solids concentration in flotation tails CIL circuit without exceeding max. slurry feed flow rate of 1,941 m3/hr	E
	Underflow Solids Concentration Range		% w/w		50-62	50-62	50-62	50-62	1	Owner	23		G
	Testwork Flux Rate		$t/h/m^2$		1.50	1.50	1.50	1.50	1	Owner	23		G
	Operating Flux Rate Range		$t/h/m^2$		1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1	Owner	16		E
	Nominal Flux Rate		$t/h/m^2$		1.09	1.01	1.08	1.21	6	Calculated			
	Design Flux Rate		$t/h/m^2$		1.30	1.21	1.29	1.21	6	Calculated		20% DF applied to Nominal Throughput	
	Thickener Area Calculated		m^2		2,827	2,827	2,827	2,827	6	Calculated			
	Diameter Selected		m		60	60	60	60	1	Owner	16		A
	Thickener Spray Water		m^3/h		0	0	0	0	2	EPC			E
	Thickener Underflow Destination(s)				CIL4 Final Tailings Tank (to CIL 2, CIL3 and TSF)	CIL4 Final Tailings Tank (to CIL 2, CIL3 and TSF)	CIL4 Final Tailings Tank (to CIL 2, CIL3 and TSF)	CIL4 Final Tailings Tank (to CIL 2, CIL3 and TSF)	1	Owner	23	Excess flow goes to the TSF	G
	Number of Underflow Discharge Pumps				2	2	2	2	1	Owner	16		E
	Pump Configuration				Duty/Standby	Duty/Standby	Duty/Standby	Duty/Standby	1	Owner	43		K
	Total Thickener Underflow		m3/hr		3320	3085	3284	3676	6	Calculated			
	Thickener Underflow - Balance to CIL 2/3		m3/hr		1831	1828	1828	1803	6	Calculated		Thickener UF reports to CIL 4 Final Tailings Tank before pumping to CIL2/3	
	Thickener Underflow - Balance to TSF (excess)		m3/hr		1491	1260	1459	1876	6	Calculated			
12.0	**CIL - FLOTATION TAILINGS**												
12.1	**General Circuit (Brownfields)**				Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3					
	Feed Streams				CIL 1 Tails and Flotation Tailings Thickener U/F	CIL 1 Tails and Flotation Tailings Thickener U/F	CIL 1 Tails and Flotation Tailings Thickener U/F	CIL 1 Tails and Flotation Tailings Thickener U/F	1	Owner	24		G
	Number of Trains				2	2	2	2	1	Owner	16		E
	Total Slurry Feed Flow Rate to CIL 2 & 3	Maximum	m^3/h		1,941	1,941	1,941	1,941	1	Owner	24		G
	Target Feed Pulp Density to CIL 2 & 3	Target	% w/w		55	55	55	55	1	Owner	16	Post dilution with CIL 1 tails and elution eluate	E
	Maximum Solids Feed Rate to CIL 2 & 3	@55% solids	t/h		1,750	1,750	1,750	1,750	1	Owner	24	From Flotation Tailings Thickener U/F	G
	CIL 1 Tails to CIL 2 & 3		t/h		43	45	45	61	6	Calculated			
	CIL 1 Tails to CIL 2 & 3		m^3/h		85	89	89	114	6	Calculated			
	Lime Solids to CIL 2 & 3		t/h		5.55	5.56	5.56	5.54	6	Calculated			
	Total Dilution to CIL 2 & 3		m3/h		24	24	24	24	6	Calculated		Includes cyanide solution, milk of lime and dilution water	
	Flotation Tailings Thickener U/F to CIL 2 & 3		t/h		1,693	1,692	1,692	1,671					
	Flotation Tailings Thickener U/F to CIL 2 & 3		% w/w		56.0	56.1	56.1	56.1	6	Calculated		Back-calculated to achieve solids concentration in flotation tails CIL circuit without exceeding max. slurry feed flow rate of 1,941 m3/hr	E
	Flotation Tailings Thickener U/F to CIL 2 & 3		m3/h		1,831	1,828	1,828	1,803	6	Calculated			
	Solids Feed Rate to CIL 2 & 3	Fresh	t/h		1,736	1,737	1,737	1,733	6	Calculated			
	Solids Feed Rate to CIL 2 & 3	Total	t/h		1,741	1,742	1,742	1,738	6	Calculated			
		per train	t/h		871	871	871	869	6	Calculated			
	Total Feed Pulp Density		% w/w		54.8	54.8	54.8	54.7	6	Calculated			
	Total Slurry Feed Flow Rate to CIL 2 & 3	Total	m^3/h		1,941	1,941	1,941	1,941	6	Calculated			
		per train	m^3/h		971	970	971	971	6	Calculated			
	CIL2/3 Leach Gold Recovery %		%		38	38	38	38	1	Owner	44	34% YTD ~~+ additional residence time~~.	0
	Adsorption Efficiency		%		88.0	88.0	88.0	88.0	1	Owner	44	Range 84 - 92%	K
12.2	**Leaching (Brownfields)**				Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3					
	Feed Streams				CIL 1 Tails and Flotation Tailings Thickener U/F	CIL 1 Tails and Flotation Tailings Thickener U/F	CIL 1 Tails and Flotation Tailings Thickener U/F	CIL 1 Tails and Flotation Tailings Thickener U/F	1	Owner	16		K
	Number of Tanks	per train			1	1	1	1	1	Owner	16		E
	Effective Tank Capacity		m^3		1,685	1,685	1,685	1,685	1	Owner	16		E
	Total Tank Capacity		m^3		1,685	1,685	1,685	1,685	6	Calculated			
	Residence Time		h		1.7	1.7	1.7	1.7	6	Calculated			
	Oxygen Addition Points				Leach Tanks and First Two Adsorption Tanks	Leach Tanks and First Two Adsorption Tanks	Leach Tanks and First Two Adsorption Tanks	Leach Tanks and First Two Adsorption Tanks	1	Owner	16		E
	Oxygen Addition Method				Side Lances in Leach Tanks and 2 Adsorption Tanks Oxygen Circulating Pump on First Leach Tank	Side Lances in Leach Tanks and 2 Adsorption Tanks Oxygen Circulating Pump on First Leach Tank	Side Lances in Leach Tanks and 2 Adsorption Tanks Oxygen Circulating Pump on First Leach Tank	Side Lances in Leach Tanks and 2 Adsorption Tanks Oxygen Circulating Pump on First Leach Tank	1	Owner	16		E
12.3	**Adsorption (Brownfields)**				Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3					
	Number of Tanks	per train			5	5	5	5	1	Owner	16		E
	Effective Tank Capacity		m^3		1,685	1,685	1,685	1,685	1	Owner	16		E
	Total Adsorption Tank Live Volume		m^3		8,425	8,425	8,425	8,425	6	Calculated			
	Residence Time		h		9	9	9	9	6	Calculated			
	Total Leach & Adsorption Time		h		10.4	10.4	10.4	10.4	6	Calculated			
	Carbon Concentration	per tank	g/L		20	20	20	20	1	Owner	16		E
	Carbon In Circuit	Total per train	t		168.5	168.5	168.5	168.5	6	Calculated			
	Design Carbon Loading		g/t		461	461	461	461	1	Owner	16		A
	Carbon Loading Range		g/t		332 - 579	332 - 579	332 - 579	332 - 579	1	Owner	16		A
	Carbon Transfer Method				Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	1	Owner	16		E
	Carbon Transfer Period		h		12	12	12	12	1	Owner	16		E
	Carbon and Pulp Transfer Rate		m^3/h		79	79	79	79	1	Owner	105		1
	Loaded Carbon Advance Method				Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	1	Owner	16		A
	Loaded Carbon Pump & Pulp Forwarding Rate		m^3/h		138	138	138	138	6	Calculated	44	Based on one 11 t strip per day. Client advice to calculate as per Ref 44	
	Loaded Carbon Transfer Time		h		4	4	4	4	1	Owner			1
12.4	**Loaded Carbon Screen (Brownfields)**				Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3					
	Feed Streams				First Adsorption Tank	First Adsorption Tank	First Adsorption Tank	First Adsorption Tank	1	Owner	16		E
	Type				Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	1	Owner	16		A
	Number	per train	No.		1	1	1	1	1	Owner	16		A
	Feed Rate		m^3/h		138	138	138	138	6	Calculated			
	Specific Screen Flux Rate	Max Recommended	$m^3/m^2/h$		35	35	35	35	1	Owner	16		E
	Minimum Area		m^2		3.9	3.9	3.9	3.9	6	Calculated			
	CIL 2 Circuit												
	Selected Size	Width	m		1.2	1.2	1.2	1.2	1	Owner	16		E
		Length	m		3.6	3.6	3.6	3.6	1	Owner	16		E
	Area		m^2		4.3	4.3	4.3	4.3	6	Calculated			
	Specific Screen Flux Rate	Actual	$m^3/m^2/h$		31.8	31.8	31.8	31.8	6	Calculated			
	Deck Type				Polyurethane	Polyurethane	Polyurethane	Polyurethane	1	Owner	16		A
	Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		A
	CIL 3 Circuit												
	Selected Size	Width	m		1.2	1.2	1.2	1.2	1	Owner	16		
		Length	m		2.4	2.4	2.4	2.4	1	Owner	16		
	Area		m^2		2.9	2.9	2.9	2.9	6	Calculated			
	Specific Screen Flux Rate	Actual	$m^3/m^2/h$		47.7	47.7	47.7	47.7	6	Calculated			
	Deck Type				Polyurethane	Polyurethane	Polyurethane	Polyurethane	1	Owner	16		
	Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		
	Total Spray Water Addition		m^3/h		10	10	10	10	1	Owner	44	Range given 1-10m3/hr	K
	Loaded Carbon Moisture		% w/w		40	40	40	40	1	Owner	4		A
	Loaded Carbon Transfer Moisture		% w/w		55	55	55	55	1	Owner	4	From Transfer Vessel	1
12.5	**Interstage Carbon Screens (Brownfields)**				Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3	Existing CIL 2 & CIL 3					
	Type				Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	1	Owner	16		E
	Number	per tank	No.		1	1	1	1	1	Owner	16		A
	Screen Feed Rate - per Screen	w/o carbon advance	m^3/h		971	970	971	971	6	Calculated			
		w/ carbon advance	m^3/h		1,050	1,049	1,050	1,050	6	Calculated			
	Area Selected	Selected	m^2		12	12	12	12	1	Owner	16		E
	Specific Screen Flux Rate	w/ carbon advance	$m^3/m^2/h$		87.5	87.5	87.5	87.5	6	Calculated			
		Suggested	$m^3/m^2/h$		78	78	78	78	9	Agreed	8	At 55%w/w solids	C
		Maximum	$m^3/m^2/h$		100	100	100	100	1	Owner	16		A
	Screen Material				Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	1	Owner	16		A
	Screen Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		A
12.6	**Carbon Safety Screen (Brownfields)**				Existing	Existing	Existing	Existing					
	Type				Delkor Linear Screens	Delkor Linear Screens	Delkor Linear Screens	Delkor Linear Screens	1	Owner	16		E
	Number		No.		4	4	4	4	1	Owner	16		E
	Area - per Screen		m^2		20	20	20	20	1	Owner	16		E
	Aperture		mm		0.85	0.85	0.85	0.85	1	Owner	16		E
	Screen Spray Water		m^3/h		5	5	5	5	1	Owner	44	Range given 1-10m3/hr	K

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
12.7	**Fine Carbon Oversize Dewatering Screen (Brownfields)**				Existing	Existing	Existing	Existing					
	Type				Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	1	Owner	16		E
	Number		No.		2	2	2	2	1	Owner	16		E
	Selected Size	Width	m		0.6	0.6	0.6	0.6	1	Owner	16		E
		Length	m		1.8	1.8	1.8	1.8	1	Owner	16		E
	Area - per Screen		m^2		1.1	1.1	1.1	1.1	1	Owner	16		E
	Aperture		mm		0.85	0.85	0.85	0.85	1	Owner	16		E
	Screen Spray Water		m^3/h		5	5	5	5	1	Owner	44	Range given 1-10m3/hr	K
12.8	**Tailings Hopper (Brownfields)**												
	Feed Streams				CIL 2 & CIL 3 Tails	CIL 2 & CIL 3 Tails	CIL 2 & CIL 3 Tails	CIL 2 & CIL 3 Tails	1	Owner	43		A
	Solids Feed Rate		t/h		1,736	1,737	1,737	1,733	6	Calculated			
	Volumetric Feed Rate		m^3/h		1,946	1,946	1,946	1,946	6	Calculated			
	Feed Solids Concentration		% w/w		54.6	54.6	54.6	54.4	6	Calculated			
	Residence Time		sec		67	67	67	67	6	Calculated			
	Tank Capacity		m^3		36	36	36	36	1	Owner	16		E
12.9	**CIL Tailings System (Brownfields)**				By Others	By Others	By Others	By Others	1	Owner	69		0
	Number of Pumps				4	4	4	4	1	Owner	16	By Others	0
	Arrangement				Duty / Duty / Duty / Standby	Duty / Duty / Duty / Standby	Duty / Duty / Duty / Standby	Duty / Duty / Duty / Standby	1	Owner	16	By Others	0
13.0	**ELUTION - COMBINED FLOTATION CONCENTRATE & FLOTATION TAIL**												
13.1	**General**				Existing Elution Circuit 1 / 2								
	Operating Schedule	Nominal	days/week		6				1	Owner	67		0
	Strips per Day	Circuit Capacity			1.5				1	Owner	16	Elution circuits can average 1.5 strips/column/day	I
	Strips per Week	Operating Range			8 - 24				1	Owner	16	Existing Operating Envelope	I
	Column Capacity	each	t		11				1	Owner	16		A
	Number of Elution Columns				2				1	Owner	16		E
	Total Circuit Capacity	total	t		22				6	Calculated			
	Dry Carbon Bulk Density		t/m^3		0.47				8	Industry			I
	Carbon Bed Volumes	per column	m^3		23.4				6	Calculated			
	Number of BVs Lean Eluate	per column			5.0				9	Agreed	8		I
	Volume of Lean Eluate	per day per column	m^3/day		175.5				6	Calculated			
		total per day	m^3/day		351.1				6	Calculated			
	Volume of Lean Eluate to CIL1		m3/hr		26				1	Owner	44		0
	Loaded Carbon Gold Grade		g/t		1,154				1	Owner	16		E
	Barren Carbon Gold Grade		g/t		100				1	Owner	16		E
	Barren Eluate Grade	Operating Range	g/t		3-8				1	Owner	23		G
	Loaded Carbon Mercury Grade	Nominal	Hg ppm		124				1	Owner	16		E
	Barren Carbon Mercury Grade	Nominal	Hg ppm		79				1	Owner	16		E
	Regenerated Carbon Mercury Grade	Nominal	Hg ppm		4				1	Owner	16		E
13.2	**Loaded Carbon Transfer (Brownfields)**												
	Carbon Transfer Time		min		180				1	Owner		3 hour transfers from existing 5.5 t transfer vessels in each CIL 1, 2 & 3 circuits	1
	Transfer Water Type				Scheme Water				1	Owner	16		A
13.3	**Carbon Rinse Screen (Brownfields)**				Existing								
	Feed Stream				CIL3 Loaded Carbon Transfer Vessel				1	Owner	43		K
	Type				Decline Vibrating Screen				1	Owner	16		A
	Number	per elution circuit	No.		1				1	Owner	16		E
	Selected Size	Width	m		1.2				1	Owner	16		E
		Length	m		2.4				1	Owner	16		E
	Area		m^2		2.9				6	Calculated			
	% Mass Split to Oversize		%		100				2	EPC			1
	Oversize Moisture %		% w/w		20				1	Owner	57		1
13.4	**Acid Washing (Brownfields)**				Existing								
	Feed Stream				Carbon Rinse Screen Oversize				1	Owner	43		E
	Number of Columns	per elution circuit			1				1	Owner	16		E
	Acid Type				Hydrochloric Acid				1	Owner	16		A
	Acid Concentration		% w/v HCl		3.0				1	Owner	16		A
	Acid Injection Time		min		17				1	Owner	16		A
	Acid Wash Solution Flowrate		BV/hr		2				6	Calculated			
	Acid Soak Time		min		60				1	Owner	16		A
	Acid Water Rinse Time		min		120				1	Owner	16		E
	Acid Rinse Bed Volumes		BV		4				1	Owner	16		1
	Acid Rinse Bed Volumes per hour		BV/hr		2				6	Calculated			
	Waste Acid Solution Destination				Final Tailings Collection Tank				1	Owner	16		E
	Acid Wash Temperature		°C		Ambient				1	Owner	44		0
13.5	**Elution (Brownfields)**												
	Number of Columns	per elution circuit			1				1	Owner	16		E
	Elution Type				Split AARL				1	Owner	16		E
	Pre-Soak Solution Strength		% w/v NaCN		1.7				1	Owner	16		A
			% w/v NaOH		2				1	Owner	16		A
13.6	**Pregnant & Lean Eluate Tanks (Brownfields)**												
	Number of Pregnant Eluate Tanks				2				1	Owner	16		E
	Pregnant Eluate Tank Capacity		m^3		200				1	Owner	16		E
	Number of Lean Eluate Tanks	per elution circuit			1				1	Owner	16		E
	Lean Eluate Tank Capacity		m^3		200				1	Owner	16		E
13.7	**Elution Electrowinning (Brownfields)**				Existing Circuits								
	Number of Circuits				2				1	Owner	16		A
	Number of Cells per Circuit	Circuit #1			2				1	Owner	16		A
		Circuit #2			3				1	Owner	16		E
	Cathode Dimensions		m x m		1.00 x 1.00				1	Owner	106		1
	Effective Cross Sectional Area	Each Cathode	m^2		1.08				6	Calculated			
	Cathode Type				Stainless Steel				1	Owner	16		A
	Cathode Wire Diameter		µm		152				1	Owner	16		E
	Number Cathodes per Cell		No.		18				1	Owner	106		1
	Rectifier Size Installed	Circuit #1	A		2,550				1	Owner	106		1
		Circuit #2	A		2,550				1	Owner	106		1
	Electrowinning Time	Circuit #1	h		12				1	Owner	16		E
		Circuit #2	h		8				1	Owner	16		E
	Eluate Circulation Tank Volume		m^3		200				1	Owner	16		A
	Barren Eluate Tank Volume		m^3		200				1	Owner	16		E
14.0	**GOLDROOM**												
14.1	**Goldroom Operation**				Existing								
	Sludge Filter Operating Days		days/week		2				1	Owner	16		E
	Sludge Dewatering Cone Capacity		m^3		1.3				1	Owner	16		E
	Sludge Filter Make				TEFSA				1	Owner	16		E
	Sludge Filter Model				HPLE				1	Owner	16	470x470mm plate filter, 6 bar design pressure	E
	Mercury Retort Oven Operating Days		days/week		2				1	Owner	16		E
	Mercury Retort Oven Make				Locchead				1	Owner	16		E
	Mercury Retort Oven Model				F66OV66				1	Owner	16	20 ft^3 electric powered mercury retort	E
	Calcine per week		kg		800 - 1,200				1	Owner	16	Existing, Calcine logs, Max Capacity 1,600 kg per cycle	E
	Smelting Furnace Operating Days		days/week		2				1	Owner	16		E
	Furnace Make				Furnace Engineering				1	Owner	16		E
	Furnace Model				Westofen KBU 80/H				1	Owner	16	2000 L x 3000 D x 1710 H, Weishaupt G3-1a burner	E
	Furnace Extraction Fan				75 kW / 1,480 RPM				1	Owner	16	IP55, 415V with 315x5SPB motor pulley and 212x5SPB fan pulley	E
	Furnace Scrubber				Metalcraft SN50247/1				1	Owner	16		E
	Smelts per week				6				1	Owner	16	Two downloads per week, three pours per day	E

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
15.0	**CARBON REGENERATION - FLOTATION CONCENTRATE & FLOTATION TAIL**												
15.1	**Barren Carbon Transfer**				**Existing**								
	Carbon Transfer Time		min		**30**				1	Owner	16		A
	Transfer Water Source				**Carbon Transfer (Scheme Water)**				1	Owner	16		E
15.2	**Regeneration Kiln Dewatering Screens (Brownfields)**				**Existing**								
	Type				**Horizontal Vibrating Screen**				1	Owner	16		E
	Number		No.		**3**				1	Owner	16	1x Per Regeneration Kiln	E
	Specific Screen Flux Rate	Max Recommended	$t/m^2/h$		**4**				1	Owner	16		E
	Selected Size	Width	m		**1.2**				1	Owner	16		E
		Length	m		**2.4**				1	Owner	16		E
	Area		m^2		2.9				6	Calculated			
	Feed % Solids		% w/w		**20**				1	Owner	57		1
	% Mass Split to Oversize		%		**100**				2	EPC			1
	Oversize Moisture %		% w/w		**60**				1	Owner	57		1
15.3	**Carbon Regeneration Kiln (Brownfields)**				**Existing**								
	Regeneration Kiln Type				**Horizontal**				1	Owner	16		E
	Regeneration Kiln Model				**Metso RK600/650HR**				1	Owner	16		E
	Number				**3**				1	Owner	16		E
	Eluted Carbon to be Regenerated		%		**100**				1	Owner	16		E
	Temperature		°C		**750**				1	Owner	16		E
	Carbon Quench Tank - Carbon Transfer Time	Each	min		**30**				1	Owner	16, 103		1
	Carbon Transfer Concentration		% w/w		**20**				1	Owner	57		1
15.4	**Regeneration Kiln Off-Gas Treatment (Brownfields)**												
	Venturi Scrubber				**Existing**								
	Number				**1**				1	Owner	16		E
	Design Dust Load		mg/Nm^3		**375**				1	Owner	16		E
	Design Mercury		g/h		**546**				1	Owner	16		E
	Design VOC		g/h		**831**				1	Owner	16		E
	Design Gas Temperature		°C		**250**				1	Owner	16		E
	Water Type				**Process**				1	Owner	16		E
	Packed Bed Wet Scrubber				**Existing**								
	Number				**1**				1	Owner	16		E
	Design Gas Flowrate		Nm^3/h		**6,481**				1	Owner	16		E
	Design Gas Temperature		°C		**80**				1	Owner	16		E
	Water Type				**Process**				1	Owner	16		E
	Mist Eliminator				**Existing**								
	Number				**1**				1	Owner	16		E
	Design Gas Flowrate		Nm^3/h		**3,916**				1	Owner	16		E
	Design Gas Temperature		°C		**58**				1	Owner	16		E
	Water Type				**Scheme**				1	Owner	16		E
	Regenerative Thermal Oxidiser				**Existing**								
	Number				**1**				1	Owner	16		E
	Design Gas Flowrate		Nm^3/h		**4,132**				1	Owner	16		E
	Design Gas Temperature		°C		**58**				1	Owner	16		E
	Carbon Filters				**Existing**								
	Number				**2 (Duty / Standby)**				1	Owner	16		E
	Design Gas Flowrate		Nm^3/h		**4,503**				1	Owner	16		E
	Design Gas Temperature		°C		**80**				1	Owner	16		E
	Design Mercury Loading		g/h		**387**				1	Owner	16		E
	Stack Emission Limits												
	Mercury		ppmv		**0.2**				1	Owner	16	FIM ERP Commissioning Report	E
	VOC		ppmv		**3.2**				1	Owner	16		E
16.0	**CYANIDE DESTRUCT**												
16.1	Feed Streams	1			**Kaltails Decant**	**Kaltails Decant**	**Kaltails Decant**	**Kaltails Decant**	1	Owner	35		1
		2			offsite reclaim **Decant & FIM I Decant**	offsite reclaim **Decant & FIM I Decant**	offsite reclaim **Decant & FIM I Decant**	offsite reclaim **Decant & FIM I Decant**	1	Owner	35		1
		3			**Decant Dam #3 & #4**	**Decant Dam #3 & #4**	**Decant Dam #3 & #4**	**Decant Dam #3 & #4**	1	Owner	35		1
	Decant Water Recovered from Tailings	% of Total Tailings Water	%		**49**	**49**	**49**	**49**	1	Owner	35		H
	Decant Water from Kaltails		m3/hr		**728**	**728**	**728**	**728**	1	Owner	44		K
	Total Feed Flow to Cyanide Water Tank		m3/hr		**1,912**	**1,835**	**1,901**	**2,040**	6	Calculated			
	Method				**Hydrogen Peroxide**	**Hydrogen Peroxide**	**Hydrogen Peroxide**	**Hydrogen Peroxide**	1	Owner	35		H
	Residual WAD Cyanide Target	Stage 1			**<0.5**	**<0.5**	**<0.5**	**<0.5**	1	Owner	35		H
	Dosing Points	Location			**Inline Mixers to each Decant Water line (3 off)**	**Inline Mixers to each Decant Water line (3 off)**	**Inline Mixers to each Decant Water line (3 off)**	**Inline Mixers to each Decant Water line (3 off)**	1	Owner	35		1
		Number			**3**	**3**	**3**	**3**	1	Owner	35		1
	Dosing Method				**Duty/duty/duty/standby pumps**	**Duty/duty/duty/standby pumps**	**Duty/duty/duty/standby pumps**	**Duty/duty/duty/standby pumps**	1	Owner	35	Standby pump acts as a lag pump to the duty operating pumps in the event of high peroxide demand	1
	Dosing Control Method				**Online CN_{WAD} Analyser**	**Online CN_{WAD} Analyser**	**Online CN_{WAD} Analyser**	**Online CN_{WAD} Analyser**	1	Owner	35		H
17.0	**REAGENTS**												
17.1	**Activator - Copper Sulphate**				**New**	**New**	**New**	**New**					
	Delivery Form				**Bulk Liquid**	**Bulk Liquid**	**Bulk Liquid**	**Bulk Liquid**	1	Owner	64		E
	Delivery Size per Trailer		L		**24,000**	**24,000**	**24,000**	**24,000**	5	Vendor	112		1
	Number of Trailers per delivery	Maximum			**2**	**2**	**2**	**2**	1	Owner	64		E
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Delivery System				**Compressed Air Displacement**	**Compressed Air Displacement**	**Compressed Air Displacement**	**Compressed Air Displacement**	1	Owner	64		E
	Delivery Side				**Central Rear**	**Central Rear**	**Central Rear**	**Central Rear**	1	Owner	64		0
	Solution Mix Strength (as $CuSO_4.5H_2O$)		% w/v		**25**	**25**	**25**	**25**	1	Owner	16		E
	Solution Mix Strength (as $CuSO_4$)		% w/v		**16**	**16**	**16**	**16**	1	Owner	64		E
	Supply Solution SG		t/m^3		**1.17**	**1.17**	**1.17**	**1.17**	1	Owner	16		E
	Storage Volume	Minimum	# deliveries		**2.5**	**2.5**	**2.5**	**2.5**	1	Owner	83		1
	Number of Storage Tanks				**1**	**1**	**1**	**1**	1	Owner	16		E
	Capacity per Tank		m^3		120	120	120	120	6	Calculated			
	Residence Time	Actual	d		11	12	11	10	6	Calculated		Residence time calculated using nominal consumption rates.	
	Consumption												
	SAG Mill Discharge	Nominal	g/t fresh feed to Cyclone Feed Hopper		**35**	**35**	**35**	**35**	1	Owner	23, 75	Added at Cyclone Feed Hopper	G
	Cleaner Scavenger	Nominal	g/t fresh cleaner scav feed		**10**	**10**	**10**	**10**	1	Owner	23, 75		G
	SAG Mill Discharge	Design	g/t fresh feed to Cyclone Feed Hopper		70	70	70	70	6	Calculated		Added at Cyclone Feed Hopper. 100% Design Factor for pumping.	
	Cleaner Scavenger	Design	g/t fresh cleaner scav feed		20	20	20	20	6	Calculated		100% Design Factor for pumping.	
	SAG Mill Discharge	Nominal	L/h		**440**	**414**	**440**	**489**	6	Calculated			
	Cleaner Scavenger	Nominal	L/h		**7.3**	**5.8**	**6.4**	**8.1**	6	Calculated			
	SAG Mill Discharge	Design	L/h		881	829	881	978	6	Calculated		100% Design Factor for pumping.	
	Cleaner Scavenger	Design	L/h		14.7	11.6	12.7	16.3	6	Calculated		100% Design Factor for pumping.	
	Nominal Consumption	as $CuSO_4.5H_2O$	L/h		447	420	446	497	6	Calculated			
		as $CuSO_4.5H_2O$	m^3/h		0.45	0.42	0.45	0.50	6	Calculated			
	Design Consumption	as $CuSO_4.5H_2O$	L/h		896	841	894	995	6	Calculated			
		as $CuSO_4.5H_2O$	m^3/h		0.90	0.84	0.89	0.99	6	Calculated			
17.2	**Collector - PAX**				**New**	**New**	**New**	**New**				Potassium Amyl Xanthate	
	Delivery Form				**Bulk Liquid**	**Bulk Liquid**	**Bulk Liquid**	**Bulk Liquid**	1	Owner	64		E
	Delivery Size per Trailer		L		**24,000**	**24,000**	**24,000**	**24,000**	5	Vendor	112		1
	Number of Trailers per delivery	Maximum			**2**	**2**	**2**	**2**	1	Owner	64		E
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Delivery System				**Compressed Air Displacement**	**Compressed Air Displacement**	**Compressed Air Displacement**	**Compressed Air Displacement**	1	Owner	64		E
	Delivery Side				**Central Rear**	**Central Rear**	**Central Rear**	**Central Rear**	1	Owner	64		0
	Solution Mix Strength		% w/v		**20**	**20**	**20**	**20**	1	Owner	64		E
	Supply Solution SG		t/m^3		**1.095**	**1.095**	**1.095**	**1.095**	1	Owner	23		E
	Number of Storage Tanks				**1**	**1**	**1**	**1**	1	Owner	16		E
	Capacity per Tank	Actual	m^3		**270**	**270**	**270**	**270**	6	Calculated			1
	Storage Volume	Actual	d		6.68	7.10	6.68	6.02	6	Calculated		Residence time calculated using nominal consumption rates.	
		Required	d		**7**	**7**	**7**	**7**	1	Owner	10, 16		G
	Consumption												
	Rougher Feed	Nominal	g/t fresh mill feed		**40**	**40**	**40**	**40**	1	Owner	16, 75		0
	Scavenger Feed	Nominal	g/t fresh mill feed		**35**	**35**	**35**	**35**	1	Owner	16, 75		0
	Cleaner Conditioning	Nominal	g/t fresh mill feed		**16**	**16**	**16**	**16**	1	Owner	16, 75		0
	Cleaner Scavenger	Nominal	g/t fresh mill feed		**16**	**16**	**16**	**16**	1	Owner	16, 75		0
	Rougher Feed	Nominal	L/h		629	592	629	699	6	Calculated			
	Scavenger Feed	Nominal	L/h		551	518	551	611	6	Calculated			
	Cleaner Conditioning	Nominal	L/h		252	237	252	280	6	Calculated			
	Cleaner Scavenger	Nominal	L/h		252	237	252	280	6	Calculated			
	Total Consumption	Nominal	L/h		1,683	1,585	1,683	1,869	6	Calculated			
		Nominal	m^3/h		1.68	1.58	1.68	1.87	6	Calculated			
	Consumption												
	Rougher Feed	Design	g/t fresh mill feed		80	80	80	80	6	Calculated		100% Design Factor for pumping	
	Scavenger Feed	Design	g/t fresh mill feed		70	70	70	70	6	Calculated		100% Design Factor for pumping	
	Cleaner Conditioning	Design	g/t fresh mill feed		32	32	32	32	6	Calculated		100% Design Factor for pumping	
	Cleaner Scavenger	Design	g/t fresh mill feed		32	32	32	32	6	Calculated		100% Design Factor for pumping	
	Rougher Feed	Design	L/h		1,258	1,185	1,258	1,398	6	Calculated		100% Design Factor for pumping	
	Scavenger Feed	Design	L/h		1,101	1,037	1,101	1,223	6	Calculated		100% Design Factor for pumping	
	Cleaner Conditioning	Design	L/h		503.3	474.0	503.3	559.1	6	Calculated		100% Design Factor for pumping	
	Cleaner Scavenger	Design	L/h		503.3	474.0	503.3	559.1	6	Calculated		100% Design Factor for pumping	
	Total Consumption	Design	L/h		3,366	3,170	3,366	3,739	6	Calculated			
		Design	m^3/h		3.37	3.17	3.37	3.74	6	Calculated			

AREA	PARAMETER		UNIT		DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
17.3	**Frother - Luprofroth 168**				**New**	**New**	**New**	**New**					
	Delivery Form				**Bulk Liquid**	**Bulk Liquid**	**Bulk Liquid**	**Bulk Liquid**	1	Owner	64		E
	Delivery Size per Trailer		kL		**26, 32 or 40**	**26, 32 or 40**	**26, 32 or 40**	**26, 32 or 40**	1	Owner	64		0
	Delivery Size per Trailer	Assumed	L		**40,000**	**40,000**	**40,000**	**40,000**	1	Owner	64		0
	Number of Trailers per delivery	Maximum			**1**	**1**	**1**	**1**	1	Owner	16		E
	Delivery Size		L		40,000	40,000	40,000	40,000	6	Calculated			
	Delivery System				**Pumped Transfer From Delivery Truck**	**Pumped Transfer From Delivery Truck**	**Pumped Transfer From Delivery Truck**	**Pumped Transfer From Delivery Truck**	1	Owner	64		E
	Delivery Side				**Passenger**	**Passenger**	**Passenger**	**Passenger**	1	Owner	64		0
	Solution Mix Strength		% w/w		**100**	**100**	**100**	**100**	1	Owner	64		E
	Supply Solution SG		t/m^3		**1.082**	**1.082**	**1.082**	**1.082**	1	Owner	16		E
	Number of Storage Tanks				**1**	**1**	**1**	**1**	2	EPC			E
	Capacity per Tank		m^3		**60**	**60**	**60**	**60**	1	Owner	16	Minimum storage volume at 1.5 x delivery volume	E
	Residence Time	Minimum	d		**7**	**7**	**7**	**7**	1	Owner	10		E
		Actual	d		29	30	29	26	6	Calculated		Residence time calculated using nominal consumption rates.	
	Consumption												
	Rougher Feed	Nominal	g/t fresh mill feed		**12**	**12**	**12**	**12**	1	Owner	16, 75		0
	Scavenger Feed	Nominal	g/t fresh mill feed		**6**	**6**	**6**	**6**	1	Owner	16, 75		0
	Cleaner Feed	Nominal	g/t fresh mill feed		**6**	**6**	**6**	**6**	1	Owner	44, 75		0
	Cleaner Scavenger	Nominal	g/t fresh mill feed		**6**	**6**	**6**	**6**	1	Owner	16, 75		0
	Rougher Feed	Nominal	L/h		34.9	32.9	34.9	38.8	6	Calculated			
	Scavenger Feed	Nominal	L/h		17.4	16.4	17.4	19.4	6	Calculated			
	Cleaner Feed	Nominal	L/h		17.4	16.4	17.4	19.4	6	Calculated			
	Cleaner Scavenger	Nominal	L/h		17.4	16.4	17.4	19.4	6	Calculated			
	Total Consumption	Nominal	L/h		87.2	82.1	87.2	96.9	6	Calculated			
			m^3/h		0.09	0.08	0.09	0.10	6	Calculated			
	Consumption												
	Rougher Feed	Design	g/t fresh mill feed		24	24	24	24	6	Calculated		100% Design Factor for pumping	
	Scavenger Feed	Design	g/t fresh mill feed		12	12	12	12	6	Calculated		100% Design Factor for pumping	
	Cleaner Feed	Design	g/t fresh mill feed		12	12	12	12	6	Calculated		100% Design Factor for pumping	
	Cleaner Scavenger	Design	g/t fresh mill feed		12	12	12	12	6	Calculated		100% Design Factor for pumping	
	Rougher Feed	Design	L/h		69.8	65.7	69.8	77.5	6	Calculated		100% Design Factor for pumping	
	Scavenger Feed	Design	L/h		34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping	
	Cleaner Feed	Design	L/h		34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping	
	Cleaner Scavenger	Design	L/h		34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping	
	Total Consumption	Design	L/h		174.4	164.3	174.4	193.8	6	Calculated			
			m^3/h		0.17	0.16	0.17	0.19	6	Calculated			
17.4	**Flocculant**				**New**	**New**	**New**	**New**				Adopt one Flocculant Mixing System - designed to incorporate Stage 2	
	Packaging Type				**Bulk Powder**	**Bulk Powder**	**Bulk Powder**	**Bulk Powder**	1	Owner	64	SNF FA920SH	E
	Delivery Size per Trailer		kg		**22,500**	**22,500**	**22,500**	**22,500**	1	Owner	64		E
	Number of Trailers per delivery	Maximum			**1**	**1**	**1**	**1**	1	Owner	64		E
	Delivery Size		kg		22,500	22,500	22,500	22,500	6	Calculated			
	Delivery System				**Pneumatic Bulk Tanker**	**Pneumatic Bulk Tanker**	**Pneumatic Bulk Tanker**	**Pneumatic Bulk Tanker**	1	Owner	64		E
	Delivery Side				**Passenger**	**Passenger**	**Passenger**	**Passenger**	1	Owner	64		0
	Storage Volume	Minimum	# deliveries		**2**	**2**	**2**	**2**	1	Owner	23		G
	Number of Mixing Tanks				**1**	**1**	**1**	**1**	1	Owner	16		E
	Mixing Tank Capacity		m^3		**80**	**80**	**80**	**80**	5	Vendor	120	Mixing tank upgrade from 40 to 80m3	1
	Solution Mix Strength		% w/v		**0.30**	**0.30**	**0.30**	**0.30**	1	Owner	44		K
	Mixes per Day per Tank				7	7	7	8	6	Calculated		Based on nominal Stage 2 consumption	
	Flocculant Dilution Factor				**10**	**10**	**10**	**10**	1	Owner	103		G
	Number of Storage Tanks				**1**	**1**	**1**	**1**	5	Vendor	36		E
	Storage Tank Capacity	Live Volume	m^3		**450**	**450**	**450**	**450**	5	Vendor	36	Vendor advice.	E
	Residence Time		h		19	20	19	17	6	Calculated		Based on nominal Stage 2 consumption	
	Consumption												
	Flotation Tailings Thickener	Nominal - Stage 1 & 2	g/t		**20**	**20**	**20**	**20**	1	Owner	23		G
		Nominal - Stage 1 & 2	kg/h		61.4	57.2	60.8	68.2	6	Calculated		Dry flocculant	
		Nominal - Stage 1 & 2	L/h		**20,473**	**19,054**	**20,278**	**22,738**	6	Calculated		Flocculant solution	
		Design - Stage 1 & 2	L/h		40,946	38,108	40,557	45,476	6	Calculated		100% Design Factor	
	Concentrate Pre-Leach Thickener	Nominal - Stage 2	g/t		**70**	**70**	**70**	**70**	1	Owner	23		
		Nominal - Stage 2	kg/h		3.0	3.1	3.1	4.3	6	Calculated		Dry flocculant	G
		Nominal - Stage 2	L/h		1,002	1,046	1,046	1,434	6	Calculated		Flocculant solution	
		Design - Stage 2	L/h		2,003	2,092	2,092	2,868	6	Calculated		100% Design Factor	
	Concentrate Thickener	Nominal - Stage 2	g/t		**70**	**70**	**70**	**70**	1	Owner	23		G
		Nominal - Stage 2	kg/h		5.2	7.3	7.3	5.8	6	Calculated		Dry flocculant	
		Nominal - Stage 2	L/h		1,748	2,430	2,430	1,949	6	Calculated		Flocculant solution	
		Design - Stage 2	L/h		3,496	4,860	4,860	3,897	6	Calculated		100% Design Factor	
	Total Nominal Consumption	Stage 1	g/h		61,421	57,163	60,836	68,216	6	Calculated			
		Stage 1	L/h		20,474	19,054	20,279	22,739	6	Calculated			
		Stage 1	m^3/h		20	19	20	23	6	Calculated			
		Stage 1	t/a		503	468	498	558	6	Calculated			
		Stage 2	g/h		69,670	67,591	71,265	78,363	6	Calculated			
		Stage 2	L/h		23,223	22,530	23,755	26,121	6	Calculated			
		Stage 2	m^3/h		23.2	22.5	23.8	26.1	6	Calculated			
		Stage 2	t/a		570	553	583	641	6	Calculated			
	Design Consumption - Mixing Plant	Stage 2	g/h		**95,000**	**95,000**	**95,000**	**95,000**	2	EPC		Based on 28.35Mtpa Total Consumption for Stage 2 when the Regrind Mill was still included. Regrind Mill was moved post tender phase.	0
		Stage 2	t/a		777	777	777	777	6	Calculated			
		Stage 2	m3/h		31.7	31.7	31.7	31.7	6	Calculated		Neat Concentration	
	Design Consumption - Dosing Pumps	Stage 2	L/h		46,446	45,060	47,509	52,241	6	Calculated		100% Design Factor	
		Stage 2	m^3/h		46.4	45.1	47.5	52.2	6	Calculated		100% Design Factor	
	Silo Live Capacity	Actual	t		**45**	**45**	**45**	**45**	5	Vendor	36		0
			d		27	28	26	24	6	Calculated		Based on nominal Stage 2 consumption	
17.5	**Flocculant**				**Existing**	**Existing**	**Existing**	**Existing**				Will be replaced by the new Flocculant Mixing System	
	Packaging Type				**Bulk Powder**	**Bulk Powder**	**Bulk Powder**	**Bulk Powder**	1	Owner	64	SNF FA920SH	E
	Delivery Size per Trailer		kg		**22,500**	**22,500**	**22,500**	**22,500**	1	Owner	64		E
	Number of Trailers per delivery	Maximum			**1**	**1**	**1**	**1**	1	Owner	64		E
	Delivery Size		kg		22,500	22,500	22,500	22,500	6	Calculated			
	Delivery System				**Pneumatic Tanker**	**Pneumatic Tanker**	**Pneumatic Tanker**	**Pneumatic Tanker**	1	Owner	64		E
	Delivery Side				**Passenger**	**Passenger**	**Passenger**	**Passenger**	1	Owner	64		0
	Number of Mixing Tanks				**1**	**1**	**1**	**1**	1	Owner	16		E
	Mixing Tank Capacity		m^3		**13**	**13**	**13**	**13**	1	Owner	16		E
	Solution Mix Strength		% w/v		**0.25**	**0.25**	**0.25**	**0.25**	1	Owner	16		E
	Mixes per Day per Tank				6	8	8	7	6	Calculated			
	Flocculant Dilution Factor				**10**	**10**	**10**	**10**	8	Industry			G
	Number of Storage Tanks				**1**	**1**	**1**	**1**	1	Owner	16		E
	Storage Tank Capacity		m^3		**33**	**33**	**33**	**33**	1	Owner	16		E
	Residence Time		h		10	8	8	8	6	Calculated			
	Consumption												
FIM	Concentrate Thickener (Flot. Concentrate CIL Pr	Nominal - Stage 1	g/t		**70**	**70**	**70**	**70**	1	Owner	16		E
			kg/h		3.0	3.1	3.1	4.3	6	Calculated			
			L/h		**1,202**	**1,255**	**1,255**	**1,720**	6	Calculated			
	Concentrate Thickener	Nominal - Stage 1	g/t		**70**	**70**	**70**	**70**	1	Owner	16		E
			kg/h		5.2	7.3	7.3	5.8	6	Calculated			
			L/h		**2,098**	**2,916**	**2,916**	**2,338**	6	Calculated			
	Total Consumption	Stage 1	g/h		8,249	10,428	10,429	10,147	6	Calculated			
		Stage 1	L/h		3,300	4,171	4,171	4,059	6	Calculated			
		Stage 1	m^3/h		3.3	4.2	4.2	4.1	6	Calculated			
		Stage 1	t/a		67	85	85	83	6	Calculated			
	Silo Capacity		m^3		**30**	**30**	**30**	**30**	1	Owner	16		E

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
17.6	**pH Modifier - Quicklime**				Existing	Existing	Existing	Existing					
	Type				Powder	Powder	Powder	Powder	1	Owner	64		0
	Packaging Type				Bulk	Bulk	Bulk	Bulk	1	Owner	64		E
	Delivery Size per Trailer		kg		30,000	30,000	30,000	30,000	1	Owner	64		G
	Number of Trailers Per Delivery		#		3	3	3	3	1	Owner	64		G
	Delivery Size		kg		90,000	90,000	90,000	90,000	6	Calculated			
	Delivery System				Pneumatic Tankers	Pneumatic Tankers	Pneumatic Tankers	Pneumatic Tankers	1	Owner	64		E
	Delivery Side				Passenger	Passenger	Passenger	Passenger	1	Owner	64		0
	Bulk Density		t/m^3		0.95	0.95	0.95	0.95	1	Owner	16		E
	Solids SG				3.37	3.37	3.37	3.37	1	Owner	16		E
	Water SG				1.07	1.07	1.07	1.07	1	Owner	16	Site metallurgical accounting assumption	E
	Solution Mix Strength		% w/w		20	20	20	20	1	Owner	16	Cyclone overflow density	E
	Lime Slurry Density		t/m^3		1.24	1.24	1.24	1.24	6	Calculated			
	Dry Consumption												
	Concentrate CIL - Nominal	concentrate basis	kg/t		51.2	51.2	51.2	51.2	1	Owner	23		G
	Concentrate CIL - Design	concentrate basis	kg/t		120.1	120.1	120.1	120.1	1	Owner	16		E
	Flotation Tailings CIL - Nominal	tailings CIL basis	kg/t		3.20	3.20	3.20	3.20	1	Owner	23		G
	Flotation Tailings CIL - Design	tailings CIL basis	kg/t		4.50	4.50	4.50	4.50	1	Owner	16		E
	Total Dry Consumption												
	Nominal		t/h		1.07	1.12	1.12	1.54	6	Calculated			
	Design		t/h		13.03	13.26	13.26	15.25	6	Calculated			
	Milk of Lime Consumption												
	Nominal		m^3/h		4.3	4.5	4.5	6.2	6	Calculated			
	Peak		m^3/h		52.6	53.5	53.5	61.6	6	Calculated			
	Storage Silo Capacity (Existing)	per Silo	t		400	400	400	400	1	Owner	16		E
	Number of Silos				2	2	2	2	1	Owner	16		E
	Residence Time		d		31	30	30	22	6	Calculated		Based on nominal consumption	
	Lime Slaker Type				Mill	Mill	Mill	Mill	1	Owner	16		E
	Number of Slakers				2	2	2	2	1	Owner	16		E
	Utilisation		%		30 - 40	30 - 40	30 - 40	30 - 40	1	Owner	16		E
	Make-up Water Type				Circuit & Cyanide Water	Circuit & Cyanide Water	Circuit & Cyanide Water	Circuit & Cyanide Water	1	Owner	16		E
	Milk of Lime Storage Tank Capacity	per tank	m^3		100	100	100	100	1	Owner	16		E
	Number of Storage Tanks				3	3	3	3	1	Owner	16		E
	Residence Time		h		69	66	66	48	6	Calculated		Based on nominal consumption	
17.7	**Hydrogen Peroxide**												
	Type				Liquid	Liquid	Liquid	Liquid	1	Owner	64		H
	Supply Strength		% w/v		70	70	70	70	1	Owner	64		0
	Supply Solution SG		t/m³		1.29	1.29	1.29	1.29	5	Vendor	111		1
	pH				4.3	4.3	4.3	4.3	1	Owner	44		0
	Delivery Form				Bulk, Isotainer	Bulk, Isotainer	Bulk, Isotainer	Bulk, Isotainer	1	Owner	64		0
	Delivery Size		L		24,000	24,000	24,000	24,000	1	Owner	64		0
	Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64		0
	Delivery Side				Central Top	Central Top	Central Top	Central Top	1	Owner	64		0
	No. Dosing Points				3	3	3	3	1	Owner	35		1
	Consumption Rate - Nominal	each Inline Mixer	L/hr		140	140	140	140	1	Owner	35	One dosing pump per mixer	H
		Total	L/hr		420	420	420	420	6	Calculated			
	Consumption Rate - Design	each Inline Mixer	L/hr		280	280	280	280	6	Calculated			
		Total	L/hr		840	840	840	840	6	Calculated			
	Number of Storage Tanks				1	1	1	1	2	EPC		316 SS Storage Tank	0
	Storage Tank Live Volume	Per Tank	m^3		84.35	84.35	84.35	84.35	6	Calculated		To achieve 7 days residence time (nominal) and cater for one 24kL isotainer delivery	
		Total	m^3		84	84	84	84	6	Calculated			
	Hydrogen Peroxide Storage Tank Residence time												
	Required Residence Time	Nominal	days		7	7	7	7	1	Owner	10		0
	Nominal Usage Rate		days		8	8	8	8	6	Calculated			
17.8	**Sodium Cyanide**				Existing	Existing	Existing	Existing					
	Delivery Form				Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64		K
	Delivery Size per Trailer		L		22,000	22,000	22,000	22,000	1	Owner	64		E
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Owner	64		E
	Delivery Size		L		44,000	44,000	44,000	44,000	6	Calculated			
	Delivery System				Compressed Air Displacement (AGR Spec)	Compressed Air Displacement (AGR Spec)	Compressed Air Displacement (AGR Spec)	Compressed Air Displacement (AGR Spec)	1	Owner	64		E
	Delivery Side				Central Top	Central Top	Central Top	Central Top	1	Owner	64		0
	Solution Mix Strength		% w/v		29	29	29	29	1	Owner	64		A
	Supply Solution SG		t/m³		1.19	1.19	1.19	1.19	1	Owner	16		A
	Consumption												
	Flotation Concentrate CIL	Operating Range	kg/t CIL per tank		2.0 - 8.0	2.0 - 8.0	2.0 - 8.0	2.0 - 8.0	1	Owner	23, 44	Dosed to four tanks.	K
		No. of Dose Points	#		4	4	4	4	1	Owner	23, 44	Dosed to four tanks	K
		Nominal	kg/t CIL per tank		5.38	5.38	5.38	5.38	1	Owner	44	Dosed to four tanks. Total of 21.50 kg/t as per Owner provided data.	0
		Nominal	kg/t CIL Total		21.5	21.5	21.5	21.5	6	Calculated			
		Nominal	kg/d		22,150	23,132	23,132	31,706	6	Calculated			
		Design	kg/t CIL per tank		10	10	10	10	1	Owner	23, 44	Dosed to four tanks.	K
		Design	kg/t CIL Total		40	40	40	40	6	Calculated			
		Design	kg/d		41,209	43,036	43,036	58,987	6	Calculated		of NaCN	
	Flotation Tailings CIL	Operating Range	kg/t CIL per tank		0.04 - 0.24	0.04 - 0.24	0.04 - 0.24	0.04 - 0.24	1	Owner	23		G
		No. of Dose Points	#		1	1	1	1	1	Owner	16	Dosed to one tank	0
		Nominal	kg/t CIL per tank		0.22	0.22	0.22	0.22	1	Owner	23		G
		Nominal	kg/t CIL Total		0.22	0.22	0.22	0.22	6	Calculated			
		Nominal	kg/d		9,164	9,170	9,169	9,148	6	Calculated			
		Design	kg/t CIL per tank		0.33	0.33	0.33	0.33	1	Owner	16		E
		Design	kg/t CIL Total		0.33	0.33	0.33	0.33	6	Calculated			
		Design	kg/d		13,745	13,756	13,754	13,722	6	Calculated		of NaCN	
	Elution Circuit 1	Nominal	m³/strip		1.2	1.2	1.2	1.2	1	Owner	16		E
	Elution Circuit 2	Nominal	m³/strip		1.2	1.2	1.2	1.2	1	Owner	16		E
		Nominal	kg/d		1,044	1,044	1,044	1,044	6	Calculated		of NaCN. Both elution circuits	
		Design	kg/d		2,088	2,088	2,088	2,088	6	Calculated		100% Design Factor	
	Intensive Leach	Nominal	kg/batch		230	230	230	230	5	Vendor	71	Based on CS10000 Acacia unit	0
		Nominal	kg/d		230	230	230	230	6	Calculated			
		Design	kg/d		460	460	460	460	6	Calculated		100% Design Factor	
	Total Consumption	Nominal	kg/t plant feed		0.874	0.874	0.874	0.874	1	Owner	44	Brownfields	K
	Nominal Consumption		kg/d		32,587	33,576	33,575	42,128	6	Calculated			
		NaCN solution (29%w/v)	L/hr		4,682	4,824	4,824	6,053	6	Calculated			
			kg/t plant feed		0.432	0.472	0.445	0.502	6	Calculated			
	Design Consumption	Design Margin	%		20	20	20	20	2	EPC		Design margin applied on nominal consumption	0
			kg/d		39,105	40,291	40,291	50,553	6	Calculated			
		NaCN solution (29%w/v)	L/hr		5,619	5,789	5,789	7,263	6	Calculated			
			kg/t plant feed		0.518	0.567	0.534	0.603	6	Calculated			
	Storage Tank Volume		m³		500	500	500	500	1	Owner	16		E
	Residence Time		d		4.4	4.3	4.3	3.4	6	Calculated			
	Number of Deliveries Required Per Day	Nominal			5	5	5	7	6	Calculated			
17.9	**Oxygen**				Existing								
	Flotation Tails CIL												
	Required Flow Rate Operating Range	Per Tank	Nm^3/t		0.03 - 0.15				1	Owner	16		E
	Required Flow Rate per Tank	Leach Tank	Nm^3/t		0.07				1	Owner	16		E
		Adsorption Tank	Nm^3/t		0.02				1	Owner	16		E
	Consumption Rate	Nominal	Nm^3/t mill feed		0.15				1	Owner	16		E
		Design	Nm^3/t mill feed		0.19				6	Calculated		25% margin on nominal	
	Flotation Concentrate CIL												
	Required Flow Rate per Tank	Leach Tank	Nm^3/t		1 - 5				1	Owner	16		E
		Adsorption Tank	Nm^3/t		1 - 3				1	Owner	16		E
	Consumption Rate	Nominal	Nm^3/t mill feed		0.38				1	Owner	16		E
		Design	Nm^3/t mill feed		0.48				6	Calculated		25% margin on nominal	
		Nominal	Nm^3/t conc		13				1	Owner	23, 44		K
	Total Consumption												
	Oxygen Demand	Nominal	Nm^3/t mill feed		0.53				1	Owner	16		E
		Design	Nm^3/t mill feed		0.66				6	Calculated			
		Nominal	kg/t mill feed		0.75				1	Owner	16	Density of O_2 = 1.428 kg/m^3 at 0 C° & 101.3 kPa	E
		Design	kg/t mill feed		0.87				6	Calculated		Calculated at NTP.	
	Oxygen Supply Available (Both Plants)		t/d		67				1	Owner	16		E
	Existing Oxygen Plant												
	PSA Capacity		t/d		14				1	Owner	16		E
	PSA Discharge Pressure		kPa		750				1	Owner	16		E
	Existing LOX Plant												
	Liquid Oxygen Capacity		t		40				1	Owner	16		E
	Liquid Oxygen Pressure		kPa		850				1	Owner	16		E

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
17.10	**Lead Nitrate**				Existing	Existing	Existing	Existing					
	Delivery Form				Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64		E
	Delivery Size per Trailer		L		24,000	24,000	24,000	24,000	5	Vendor	112		1
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Owner	64		E
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Preferred Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64		E
	Delivery Side				Central Rear	Central Rear	Central Rear	Central Rear	1	Owner	64		0
	Solution Mix Strength		% w/v		40	40	40	40	1	Owner	64		E
	Supply Solution SG		t/m³		1.17	1.17	1.17	1.17	1	Owner	16		E
	Consumption												
	Concentrate Leaching	Nominal	g/t CIL Feed		10,000	10,000	10,000	10,000	1	Owner	23		G
			L/h		280	280	280	280	6	Calculated			
	Concentrate Leaching	Design	g/t CIL Feed		15,000	15,000	15,000	15,000	1	Owner	16	50% addition rate design margin	E
			L/h		1,610	1,681	1,681	2,304	6	Calculated			
	Number of Storage Tanks				1	1	1	1	1	Owner	16		E
	Storage Tank Capacity		m³		60	60	60	60	1	Owner	16		E
	Residence Time		days		8.93	8.93	8.93	8.93	6	Calculated		Based on Nominal Consumption	
17.11	**Sodium Hydroxide (Caustic)**				Existing								
	Delivery Form				Bulk Liquid				1	Owner	64		E
	Delivery Size per Trailer		L		20,000				1	Owner	64		E
	Number of Trailers per delivery	Maximum			3				1	Owner	64		E
	Delivery Size		L		60,000				6	Calculated			
	Preferred Delivery System				Pumped Transfer From Truck				1	Owner	64		E
	Delivery Side				Passenger				1	Owner	64		0
	Solution Mix Strength		% w/w		50				1	Owner	64		E
	Supply Solution SG		t/m³		1.50				1	Owner	16		E
	Nominal Consumption												
	Elution Circuit 1		m³/strip		1.7				1	Owner	16		E
	Elution Circuit 2		m³/strip		1.7				1	Owner	16		E
	Intensive Leach		kg/batch		29				5	Vendor	71	Based on CS10000 Acacia unit	0
	Nominal Consumption Rate		kg/t mill feed		0.38				1	Owner	16	Estimate, dependent on elution strategy	E
			kg/t mill feed		1,195				6	Calculated		At 25.5 Mtpa	
			L/h		1,594				6	Calculated		At 25.5 Mtpa	
			m³/h		1.59				6	Calculated		At 25.5 Mtpa	
	Storage Tank Capacity		m³		70.5				1	Owner	16		E
	Residence Time		days		1.8				6	Calculated		At 25.5 Mtpa, based on nominal consumption	
17.12	**Hydrochloric Acid**				Existing								
	Delivery Form				Bulk Liquid				1	Owner	64		E
	Delivery Size per Trailer		L		20,000				1	Owner	64		E
	Number of Trailers per delivery	Maximum			3				1	Owner	64		E
	Delivery Size		L		60,000				6	Calculated			
	Preferred Delivery System				Pumped Transfer From Truck				1	Owner	64		E
	Delivery Side				Passenger				1	Owner	64		0
	Supply Strength		% w/w		32				1	Owner	64		E
	Supply Solution SG		t/m³		1.16				1	Owner	16		E
	Nominal Consumption												
	Acid Wash Column 1		m³/strip		2.7				1	Owner	16		E
	Acid Wash Column 2		m³/strip		2.7				1	Owner	16		E
			m³/d		8.1				6	Calculated			
	Nominal Consumption Rate		kg/t mill feed		0.10				1	Owner	16		E
	Storage Tank Capacity		m³		40				1	Owner	16		E
	Residence Time		days		4.9								
17.13	**Grinding Media**												
	SAG Mill												
	Number of Bunkers				2	2	2	2	1	Owner	16		E
	Bunker Storage Size		m³		35	35	35	35	1	Owner	16		E
	Media Type				Forged Steel	Forged Steel	Forged Steel	Forged Steel	1	Owner	16, 38		E
	Media Diameter		mm		125	125	125	125	1	Owner	16, 38		E
	Bulk Density		t/m³		4.65	4.65	4.65	4.65	1	Owner	16		E
	Package Form				Bulk	Bulk	Bulk	Bulk	1	Owner	16		E
	Delivery Size		t		25	25	25	25	1	Owner	23		G
	Trailers per Delivery		#		1	1	1	1	1	Owner	23		G
	Consumption Rate - Existing SAG		kg/t mill feed		0.35	0.35	0.35	0.35	1	Owner	16, 38		E
	Consumption Rate - New SAG		kg/t mill feed		0.40	0.40	0.40	0.40	1	Owner	16, 38		E
	Total Consumption		t/a		9,759	9,158	9,759	10,899	6	Calculated			
	Storage Capacity		t		325.5	325.5	325.5	325.5	6	Calculated			
	Residence Time		d		12	13	12	11	6	Calculated			
	Indexer/Charger Storage Capacity	Minimum	Buckets		1.5	1.5	1.5	1.5	1	Owner	16	CAT972 Bucket	E
	Ball Mill												
	Number of Bunkers				1	1	1	1	1	Owner	16		E
	Bunker Storage Size		m³		35	35	35	35	1	Owner	16		E
	Media Type				High Chrome	High Chrome	High Chrome	High Chrome	1	Owner	16, 38		E
	Media Diameter		mm		75	75	75	75	1	Owner	16, 38		E
	Bulk Density		t/m³		4.75	4.75	4.75	4.75	1	Owner	16		E
	Package Form				Bulk	Bulk	Bulk	Bulk	1	Owner	16		E
	Delivery Size		t		25	25	25	25	1	Owner	23		G
	Trailers per Delivery		#		1	1	1	1	1	Owner	23		G
	Consumption Rate		kg/t mill feed		0.20	0.20	0.20	0.20	1	Owner	16, 38		E
	Consumption		t/a		5,148	4,847	5,148	5,718	6	Calculated			
			kg/h		629.2	592.4	629.2	698.8	6	Calculated			
	Storage Capacity		t		166.3	166.3	166.3	166.3	6	Calculated			
	Residence Time		d		12	13	12	11	6	Calculated			
	Indexer/Charger Storage Capacity	Minimum	Buckets		1.5	1.5	1.5	1.5	1	Owner	16	CAT972 Bucket	E
	UFG Mill												
	Media Type				Ceramic	Ceramic	Ceramic	Ceramic	1	Owner	16		E
	Media Diameter		mm		2.5	2.5	2.5	2.5	1	Owner	16		E
	Bulk Density		t/m³		2.2	2.2	2.2	2.2	1	Owner	16		E
	Package Form				Bulka Bag	Bulka Bag	Bulka Bag	Bulka Bag	1	Owner	16		E
	Package Weight		kg		2,000	2,000	2,000	2,000	1	Owner	16		E
	Addition Rate		kg/t mill feed		0.56	0.56	0.56	0.56	1	Owner	10		0
		UFG Mill 1	kg/h		5.6	5.6	5.6	7.3	6	Calculated			
		UFG Mill 1	kg/day		134.4	134.4	134.4	174.7	6	Calculated			
		UFG Mill 2	kg/h		16.8	16.8	16.8	25.8	6	Calculated			
		UFG Mill 2	kg/day		403.2	403.2	403.2	618.2	6	Calculated			
	Site Storage Capacity	Minimum	# Bags		50	50	50	50	1	Owner	10		0
	Site Storage Capacity	Design	# Bags		54	54	54	54	2	EPC		Navisworks model. Designed to achieve the minimum.	0
	Residence Time		days		201	201	201	136	6	Calculated		UFG Mill 1 & 2 Consumption	
	Media Hopper Capacity	UFG Mill 2	m³		14	14	14	14	5	Vendor	65	Isacharger capacity negligible.	0
	Residence Time	UFG Mill 2	days		76.4	76.4	76.4	49.8	6	Calculated			
	Residence Time	UFG Mill 2	# Bags		15.4	15.4	15.4	15.4	6	Calculated			
	Additional Bunker (Previously the Regrind Mill Bunker)												
	Number of Bunkers				1	1	1	1	1	Owner	44		0
	Bunker Storage Size		m³		7.9	7.9	7.9	7.9	6	Calculated		Takes a full delivery x 150%	H
	Media Diameter		mm		25	25	25	25	1	Owner	38		H
	Bulk Density		t/m³		4.75	4.75	4.75	4.75	1	Owner	44		0
	Package Form				Bulk	Bulk	Bulk	Bulk	1	Owner	44		0
	Delivery Size		t		25	25	25	25	1	Owner	44		0
17.14	**Antiscalant**												
	Type				FF585				1	Owner	16		E
												New addition, to be dosed into gravity concentrator	
	Addition Rate - Gravity Concentrators		ml/min		40				1	Owner	16	fluidisation water line	E
	Addition Rate - Kiln Scrubber		ml/min		45				1	Owner	16	Existing	E
	Type				FF540				1	Owner	16		E
	Addition Rate - Elution Circuit 1		ml/min		56				1	Owner	16	Existing	E
	Addition Rate - Elution Circuit 2		ml/min		70				1	Owner	16	Existing	E
	Addition Rate - Elution Circuit 1 Heat Exchanger		ml/min		24				1	Owner	16	Existing	E
	Addition Rate - Elution Circuit 2 Heat Exchanger		ml/min		30				1	Owner	16	Existing	E
17.15	**LPG**												
	Type				Propane				1	Owner	16		E
	Consumption Rate	Plant Feed Basis	L/t		0.22				1	Owner	16		E
	Elution Burner 1 Range	High Flame	MJ/hr		4,900				1	Owner	16		E
		Low Flame	MJ/hr		2,000				1	Owner	16		E
	Elution Burner 2 Range	High Flame	MJ/hr		4,900				1	Owner	16		E
		Low Flame	MJ/hr		2,000				1	Owner	16		E
	Regeneration Kiln 1		MJ/hr		4,500				1	Owner	16		E
	Regeneration Kiln 2		MJ/hr		4,500				1	Owner	16		E
	Regenerative Thermal Oxidiser		MJ/hr		1,580				1	Owner	88		1
	Smelting Furnace	Maximum	MJ/hr		2,000				1	Owner	16		E

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35MTPA THROUGHPUT NOMINAL GRADE P80 220um					
18.0	**WATER SERVICES - EXISTING**												
18.1	**Saline Bore Water**												
	Source					Borefields			1	Owner	16		A
	Storage Method 1					Dam 1 (North)			1	Owner	16		A
	Live Volume		m^3			8,000			1	Owner	16		A
	Dam Water Pumps		#			2			1	Owner	23		G
	Pump Model					Stalker 250-250-400, 132kW			1	Owner	23		G
	Storage Method 2					Dam 2 (South)			1	Owner	16		A
	Live Volume		m^3			12,000			1	Owner	16		A
	Dam Water Pumps		#			2			1	Owner	23		G
	Pump Model					TKL 250x250 - 400HT			1	Owner	23		G
	Total Live Volume		m^3			20,000			6	Calculated			
18.2	**Mill Circuit Water**												
	Source			FIM	Concentrate Thickener O/F Concentrate Thickener O/F Circuit Water Tank	Concentrate Thickener O/F Concentrate Thickener O/F Circuit Water Tank	Concentrate Thickener O/F Concentrate Thickener O/F Circuit Water Tank	Concentrate Thickener O/F Concentrate Thickener O/F Circuit Water Tank	1	Owner	43, 66		0
	Storage Method								1	Owner	16		E
	Live Volume		m^3		1,500	1,500	1,500	1,500	1	Owner	16		E
	Residence Time		min		105.8	104.8	104.8	104.9	6	Calculated			
	Water Demand - Nominal	Calculated	m^3/h		851	859	859	858	6	Calculated			
	Water Demand - Design		m^3/h		1,800	1,800	1,800	1,800	1	Owner	16		E
	Design Flowrate per Pump		m^3/h		1,800	1,800	1,800	1,800	1	Owner	16		E
	Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	16		E
	Pump Model				APP51-300	APP51-300	APP51-300	APP51-300	1	Owner	23		G
18.3	**Cyanide Water**												
	Source					Decant Return			1	Owner	16		E
	Storage Method					Cyanide Water Tank			1	Owner	16		E
	Live Volume		m^3			1,115			1	Owner	16		E
	Cyanide Water Pump Configuration					Duty / Standby			1	Owner	16		E
	Cyanide Water Pump Model					TKL 200x200 - 500HT			1	Owner	23		G
	Cyanide Transfer Pump Configuration					Duty / Standby			1	Owner	16		E
	Cyanide Transfer Pump Model					TKL 100x65 - 315HT			1	Owner	23		G
18.4	**Scheme Water**												
	Water Demand		m^3/h		182	181	182	185	6	Calculated		Includes carbon transfer water	
	Scheme Water Quality												
	SG		-		1.00	1.00	1.00	1.00	1	Owner	1		0
	TDS		mg/L		454	454	454	454	1	Owner	1		0
	pH		-		8.0	8.0	8.0	8.0	1	Owner	1		0
	Chlorides	Range	mg/L		232	232	232	232	1	Owner	1		0
18.5	**Gland Water**												
	Source					Dam 2 (South)			1	Owner	16		E
	Storage Method					Gland Water Tank			1	Owner	16		E
	Live Volume		m^3			83			1	Owner	16		E
	Pump Configuration					Duty / Standby			1	Owner	16		E
	Pump Model					TIK 90kW			1	Owner	23		G
18.6	**Eastern Borefield Transfer Dam**												
	Source					Seepage			1	Owner	16		E
	Storage Method					Dam			1	Owner	16		E
	Destination					Dam 3			1	Owner	79		0
	Live Volume		m^3			12,000			1	Owner	16		E
	Pump Configuration					Duty / Standby			1	Owner	16		E
18.7	**Saline Gravity Dam**												
	Source					Borefield			1	Owner	16		E
	Storage Method					Dam			1	Owner	16		E
	Destination					Dam 1 (North)			1	Owner	79		0
	Live Volume		m^3			16,000			1	Owner	16		E
18.8	**Kaltails Saline Dam**												
	Source					Borefield			1	Owner	16		E
	Storage Method					Dam			1	Owner	16		E
	Destination					Dam 3			1	Owner	79		0
	Live Volume		m^3			13,600			1	Owner	16		E
	Pump Configuration					Duty / Duty / Standby			1	Owner	16		E
18.10	**Saline Transfer Dam** offsite reference												
	Source					Borefield			1	Owner	16		E
	Storage Method					Dam			1	Owner	16		E
	Destination					Dam 1 (North) and Dam 2 (South)			1	Owner	79		0
	Live Volume		m^3			340			1	Owner	16		E
	Pump Configuration					Duty / Standby			1	Owner	16		E
18.11	**Southern Surge Tank**												
	Storage Method					Southern Surge Tank			1	Owner	16		E
	Destination					Dam 2 (South)			1	Owner	16		E
	Capacity		m^3			575			1	Owner	16		E
19.0	**WATER SERVICES - NEW**												
19.1	**Process Water**												
	Source				Flotation Tailings Thickener O/F Saline Bore Water Makeup Tank	Flotation Tailings Thickener O/F Saline Bore Water Makeup Tank	Flotation Tailings Thickener O/F Saline Bore Water Makeup Tank	Flotation Tailings Thickener O/F Saline Bore Water Makeup Tank	1	Owner	10		E
	Storage Method								9	Agreed	10		E
	Live Volume		m^3		2,080	2,080	2,080	2,080	9	Agreed	10		E
	Residence Time	Minimum	min		15	15	15	15	1	Owner	16		E
	Water Demand	Calculated	m^3/h		5,155	4,922	5,258	4,390	6	Calculated			
	Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	9	Agreed	10		E
	Process Water Quality												
	SG		-		1.07	1.07	1.07	1.07	1	Owner	1		K
	TDS		mg/L		140,000	140,000	140,000	140,000	1	Owner	1		K
	pH		-		4.6	4.6	4.6	4.6	1	Owner	1		K
	Chlorides	Range	mg/L		20,492 - 86,243	20,492 - 86,243	20,492 - 86,243	20,492 - 86,243	1	Owner	1		K
19.2	**Raw Water**												
	Source (Brownfields)				Dam 2 (South)	Dam 2 (South)	Dam 2 (South)	Dam 2 (South)	1	Owner	43	Supplies the Raw Water Tank	E
	Storage Method				Steel Tank	Steel Tank	Steel Tank	Steel Tank	1	Owner	68		0
	Supply Configuration				Separate Raw, UFG 2/3 Gland, LP and HP Gland	Separate Raw, UFG 2/3 Gland, LP and HP Gland	Separate Raw, UFG 2/3 Gland, LP and HP Gland	Separate Raw, UFG 2/3 Gland, LP and HP Gland	1	Owner	68		0
	Total Live Volume	Actual	m^3		375	375	375	375	1	Owner	68		0
	Total Live Volume	Required	m^3		377	377	377	377	6	Calculated			
	Residence Time	Actual	min		60	60	60	60	6	Calculated			
		Required	min		60	60	60	60	9	Agreed	10, 16	Nominal Basis	K
	Total Nominal Water Demand		m^3/h		377	377	377	377	2	EPC	68	Inclusive of HP and LP gland water and UFG 2 & 3 gland water	0
	Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	16		E
	Source (Greenfields)				Dam 3	Dam 3	Dam 3	Dam 3	1	Owner	69	By Others	0
	Live Volume		m^3		14,000	14,000	14,000	14,000	1	Owner	69	By Others	0
19.3	**Gland Water**												
	Source				Raw Water Tank	Raw Water Tank	Raw Water Tank	Raw Water Tank	1	Owner	43		A
	Total Nominal Gland Water Demand		m^3/h		87	87	87	87	2	EPC	68	Inclusive of HP and LP gland water and UFG 2 & 3 gland water	0
	Low Pressure Gland Water Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	43		A
	High Pressure Gland Water Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	43		A
20.0	**AIR SERVICES**												
20.1	**Low Pressure Flotation Air**												
	Number of Flotation Blowers					4			1	Owner	49		I
	Operating Configuration					Duty / Duty / Duty / Standby			1	Owner	49		I
	Type					Low pressure, oil free screw blower with sound enclosure			1	Owner	49		I
	Capacity - Each	Design	Am^3/h			8,688			1	Owner	49		I
	Capacity - Total		Am^3/h			26,064			6	Calculated			
	Pressure	Design	kPag			119.7			1	Owner	49		I
20.2	**High Pressure Air**												
	Number of Compressors					3			1	Owner	43		A
	Operating Configuration					Duty / Duty / Standby			1	Owner	43		D
	Type					Oil Injected Rotary Screw Compressor			5	Vendor	45		0
	Capacity - Each	NTP Conditions	L/s			750			5	Vendor	45		D
	Pressure		kPag			980			5	Vendor	45		0
	Duty					New facilities only			2	EPC			A